

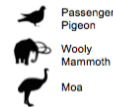
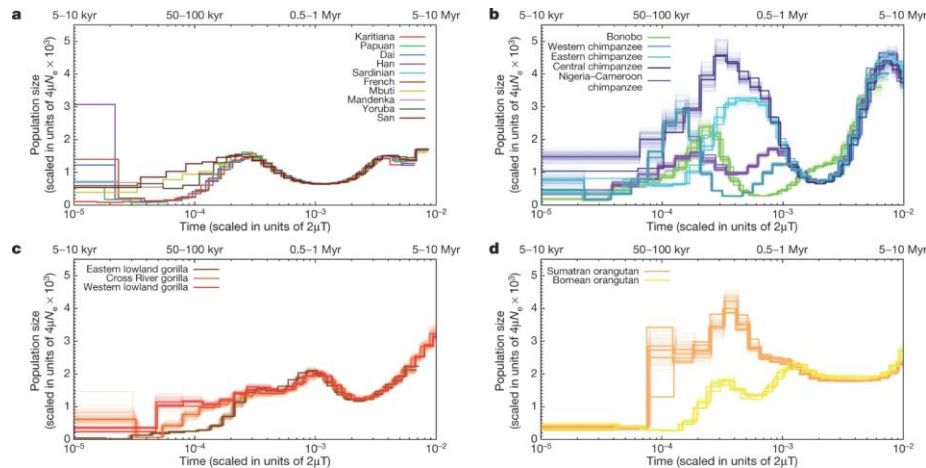
Inferring population size changes over time.
Clustering of populations using haplotype-based and
tree-based methods. Inference of coalescence rates
and effective population sizes

Oscar Lao

Overview

- Why infer $N_e(t)$?
- Coalescence and recombination
- SMC as an approximation to the ARG
- HMM intuition for local genealogies
- Main SMC-based methods
- Interpretation, assumptions, and pitfalls

Conservation genetics and estimation of effective population size over time



Extinct



Extinct in the Wild



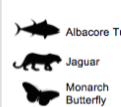
Critically Endangered



Endangered



Vulnerable



Near Threatened



Least Concern

Interpreting Effective Population Size (N_e)

- What is effective population size?
- Drift-based interpretation
- Long-term vs local-in-time N_e
- Why genetic diversity reflects coalescent history

“ N_e is the size of an ideal population that would experience the same rate of genetic drift as the population in question.”

Journal of Heredity, 2022, 113, 371–379

Interpreting Effective Population Size (N_e)

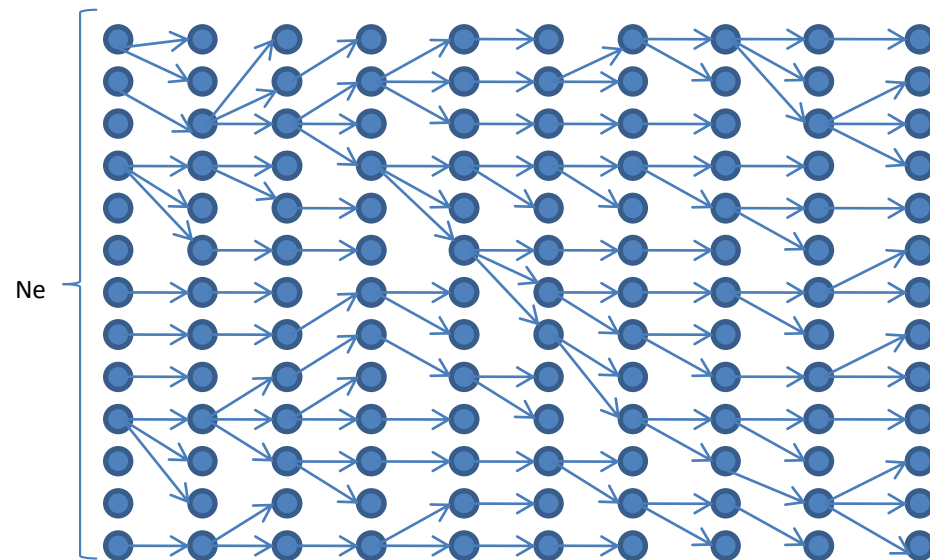
- What N_e is not
- Not census size
- Not always literal demography
- Can reflect structure, linked selection, and admixture

“Effective population size is the number of individuals that contribute offspring to the next generation.”

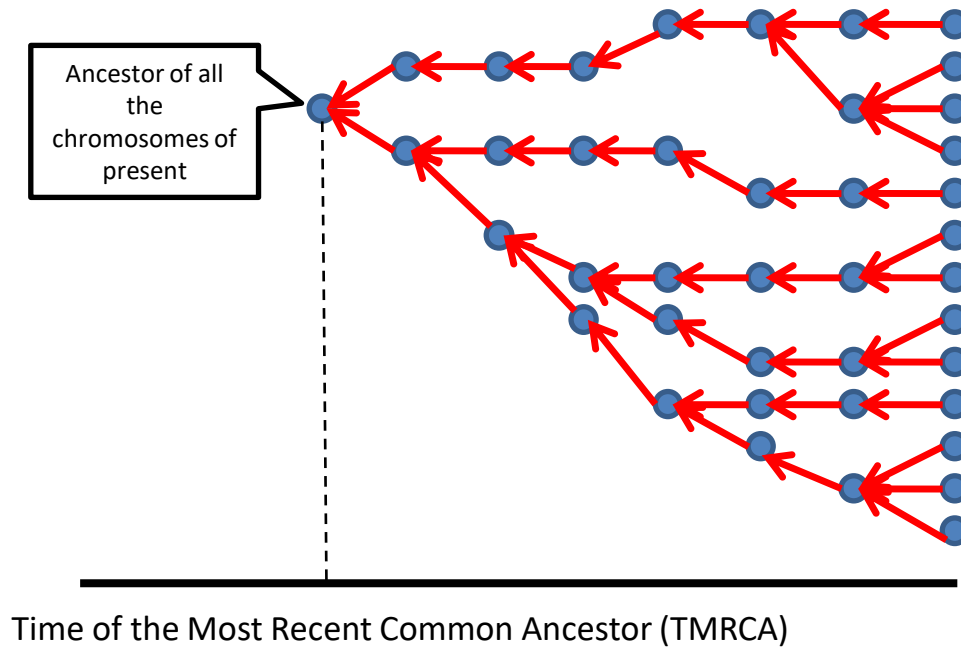
Although it is possible to find scenarios where this statement holds, it is not true in any general sense.

Journal of Heredity, 2022, 113, 371–379

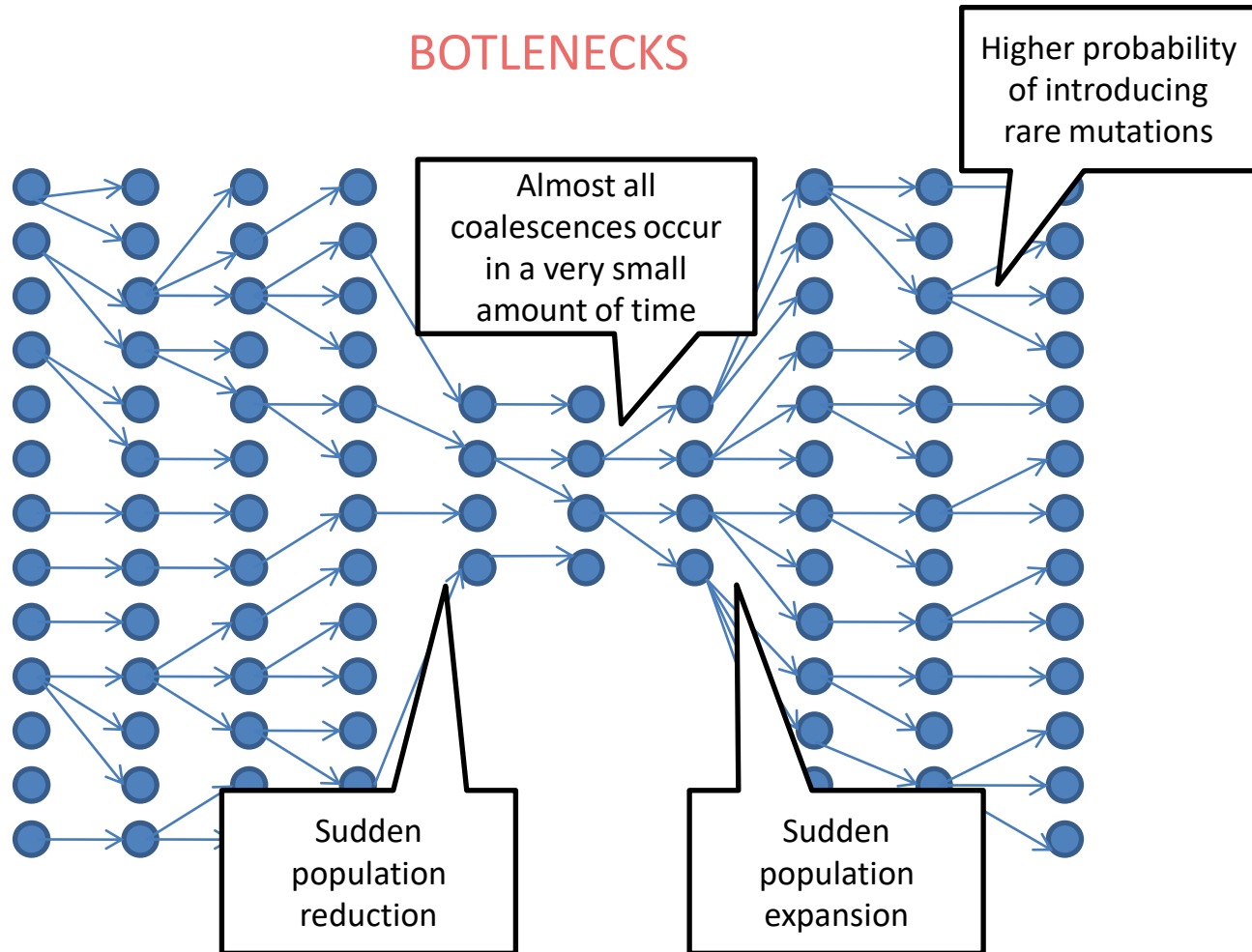
REMEMBER: ALL YOU NEED IS COALESCENCE



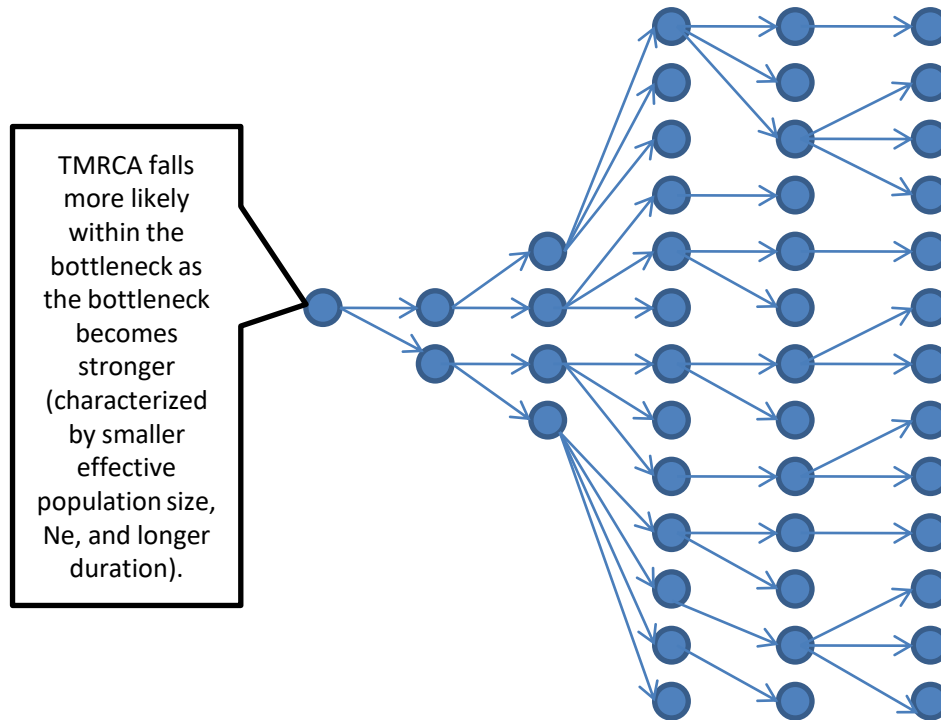
REMEMBER: ALL YOU NEED IS COALESCENCE



GOING BEYOND WRIGHT-FISHER BOTTLENECKS



GOING BEYOND WRIGHT-FISHER BOTTLENECKS



GOING BEYOND WRIGHT-FISHER BOTLENECKS

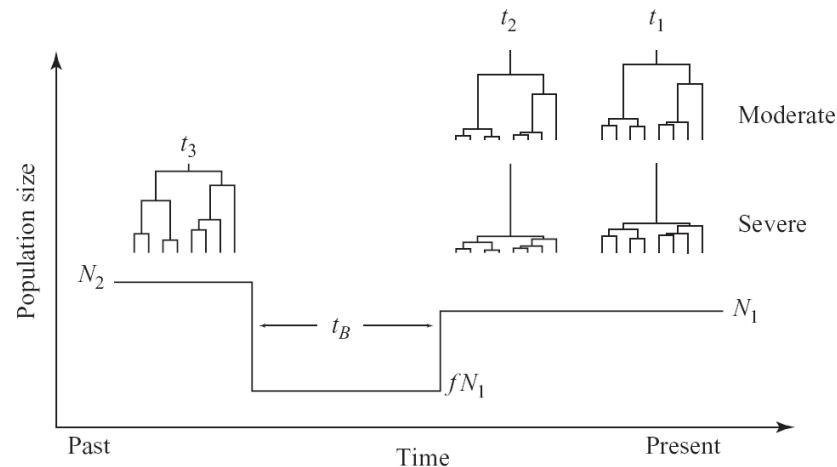
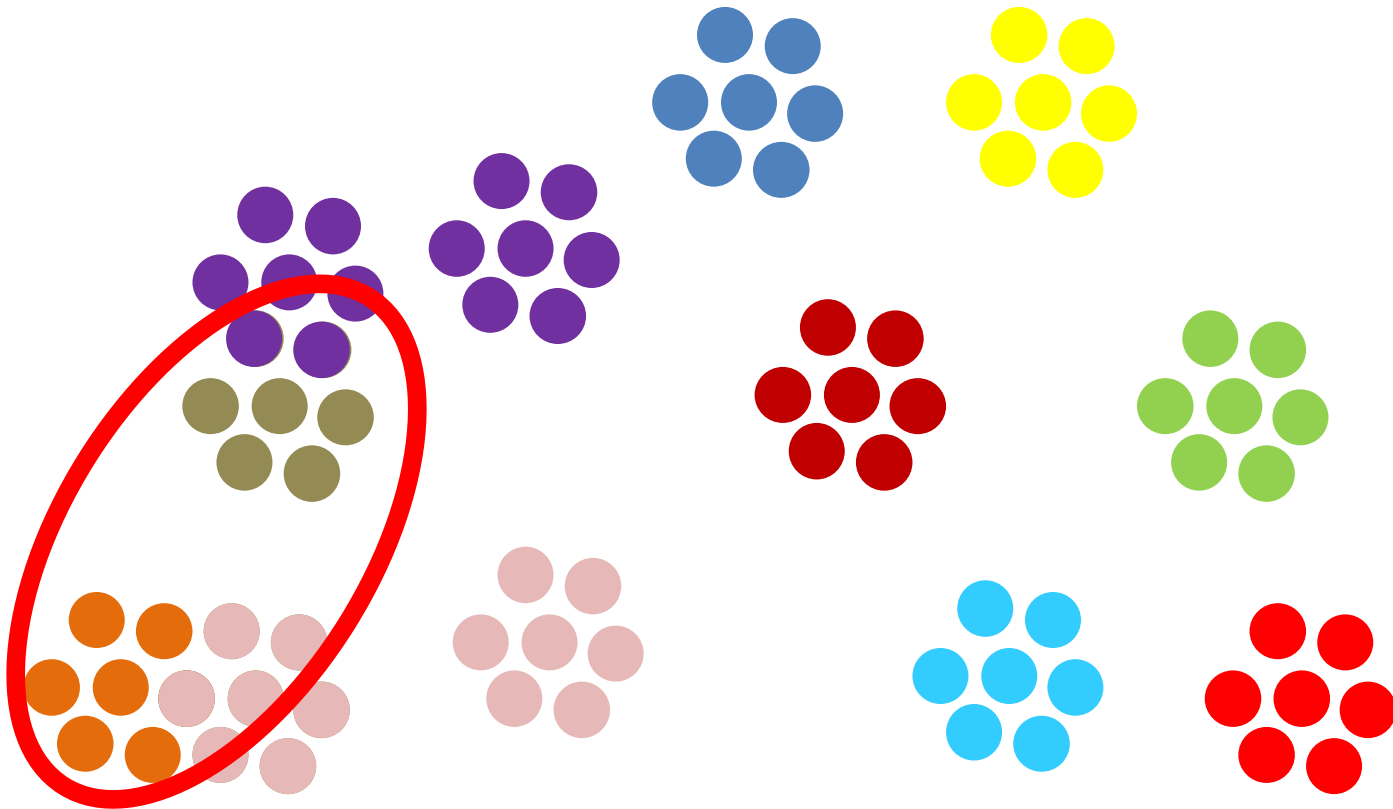
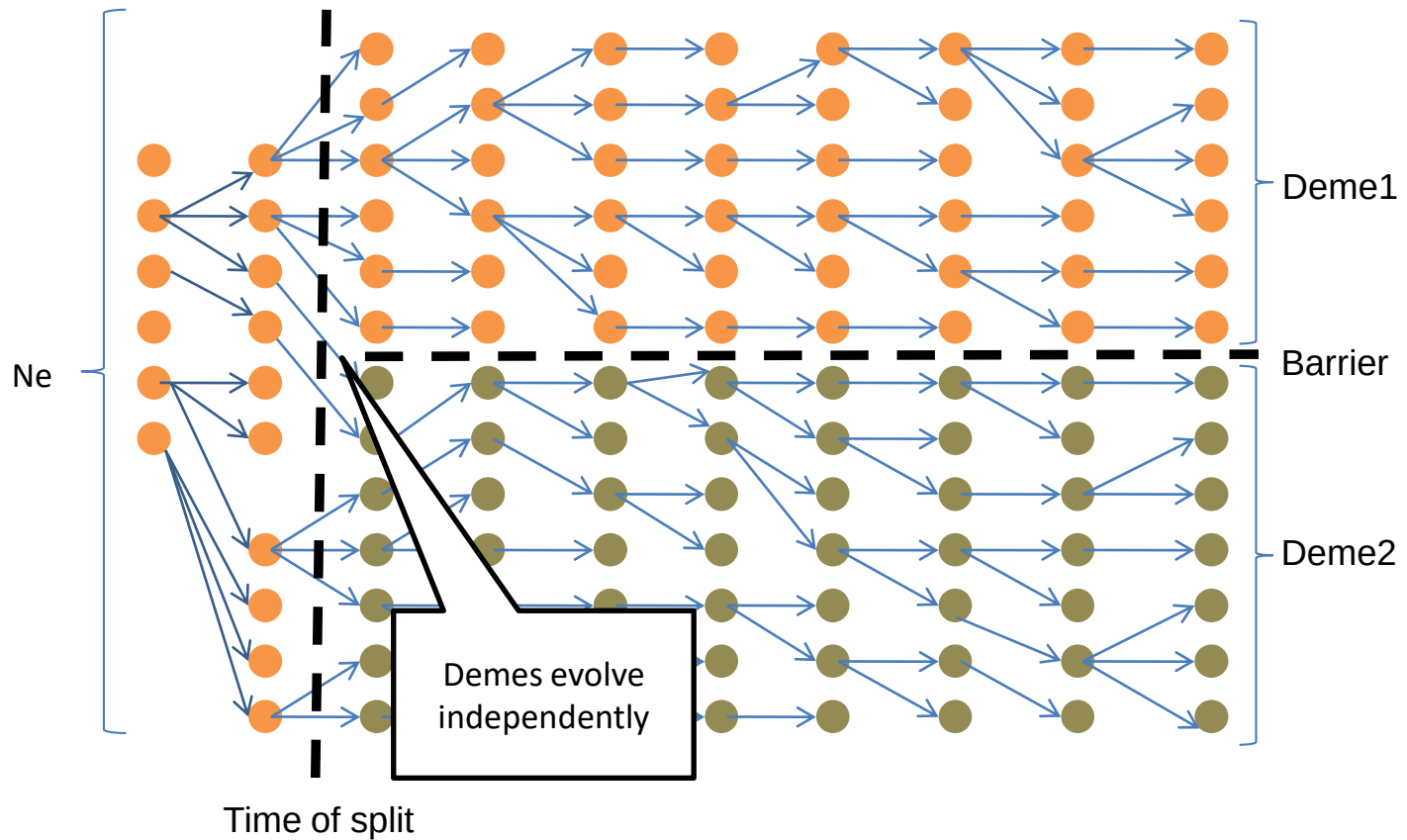


Figure 4.8 The effect of a population bottleneck on the genealogical tree of a sample. The shape of the genealogical tree depends on when the genes are sampled (or equivalently when the bottleneck occurred relative to the time of sampling). In the figure example trees are shown when sampling occurs after the bottleneck, shortly before the bottleneck and some time before the bottleneck (time is running backwards). f denotes the fraction of the current population surviving through the bottleneck and t_B is the duration of the bottleneck. Severe bottleneck: most coalescence events occur during the bottleneck. Moderate bottleneck: some lineages are likely to survive through the bottleneck and find MRCA's while the population has size N_2 .

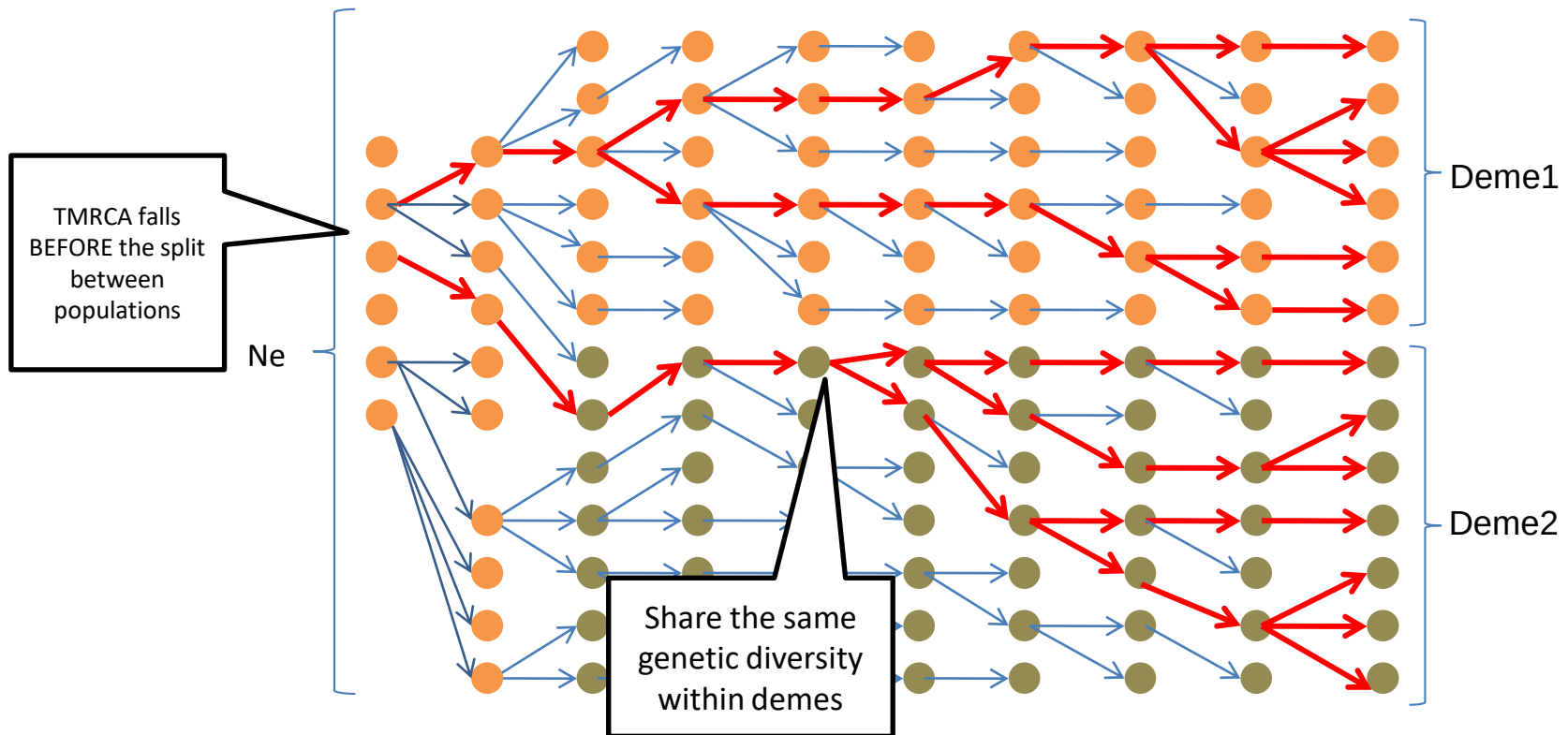
GOING BEYOND WRIGHT-FISHER POPULATION SUBSTRUCTURE



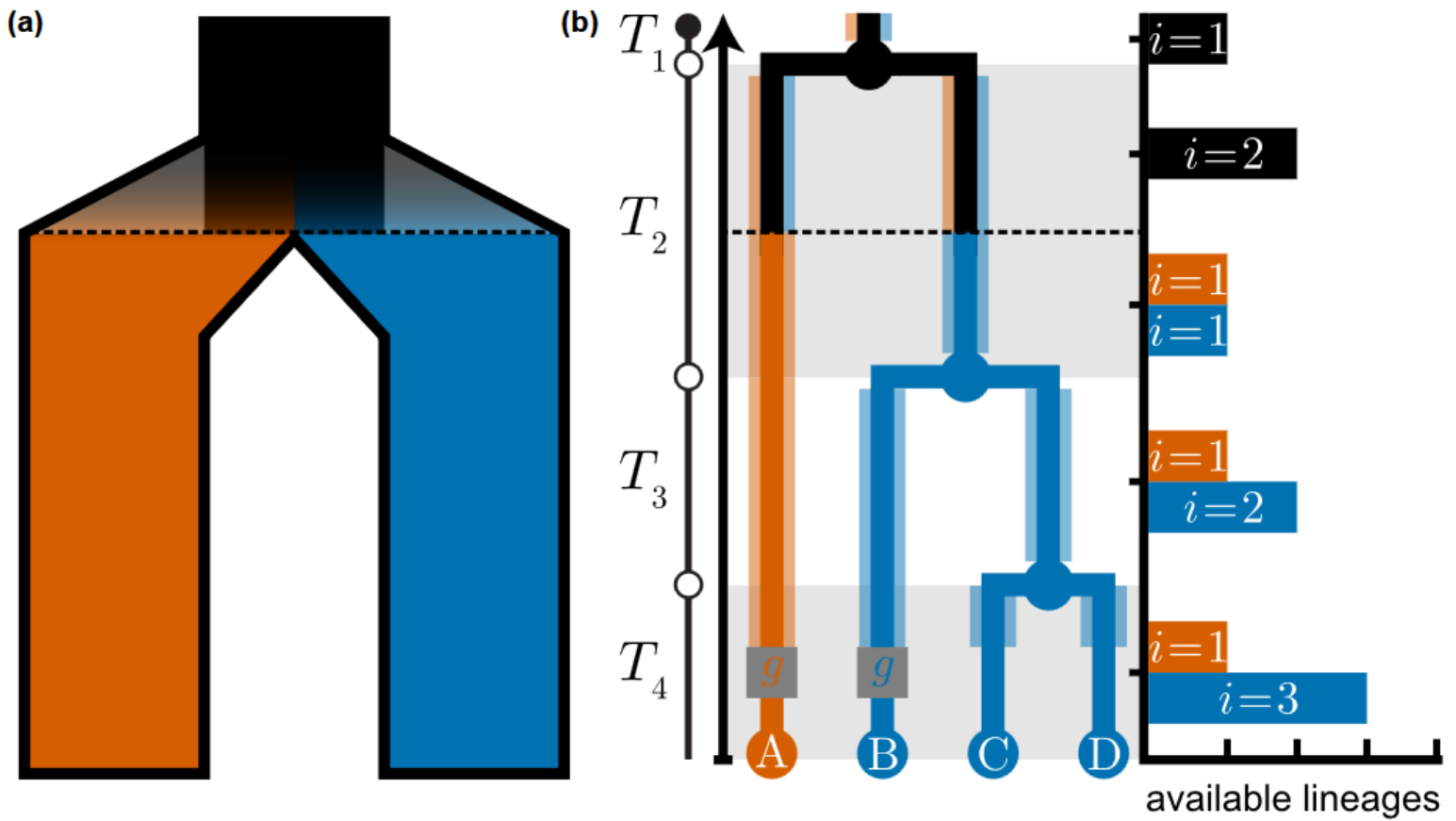
GOING BEYOND WRIGHT-FISHER POPULATION SUBSTRUCTURE AND COALESCENCE



GOING BEYOND WRIGHT-FISHER POPULATION SUBSTRUCTURE AND COALESCENCE

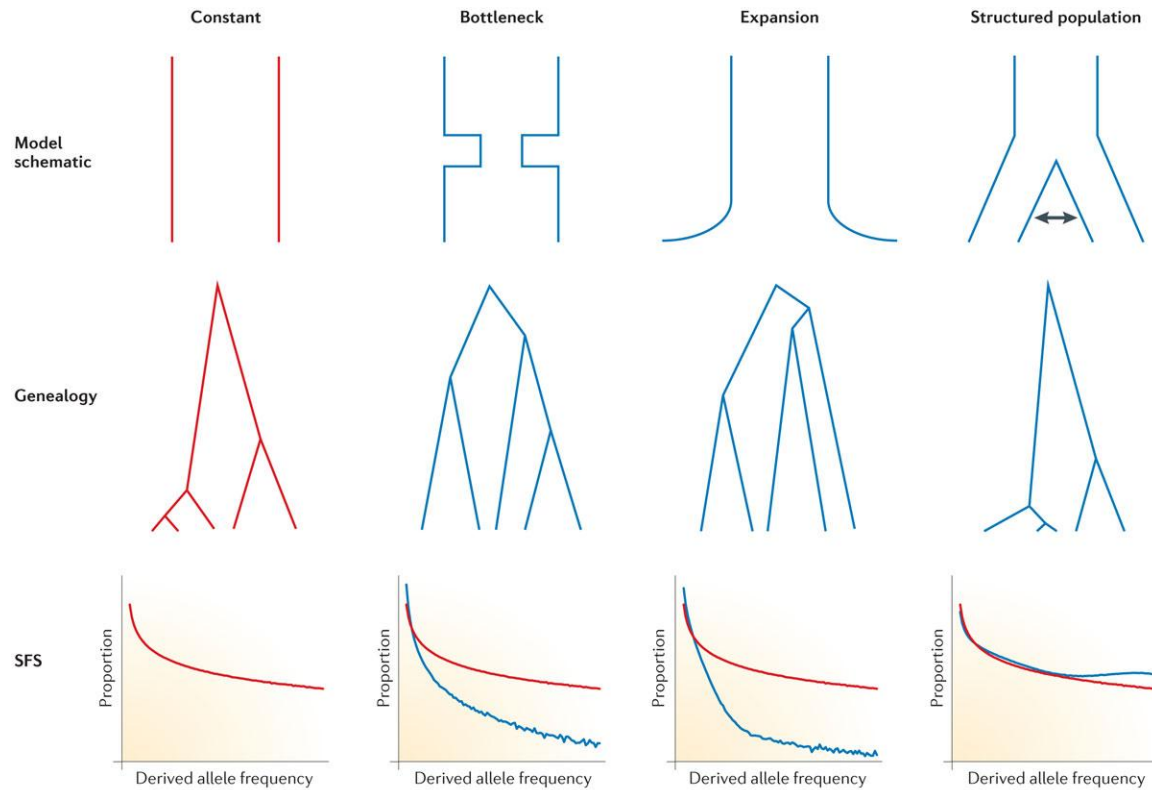


GOING BEYOND WRIGHT-FISHER POPULATION SUBSTRUCTURE AND COALESCENCE

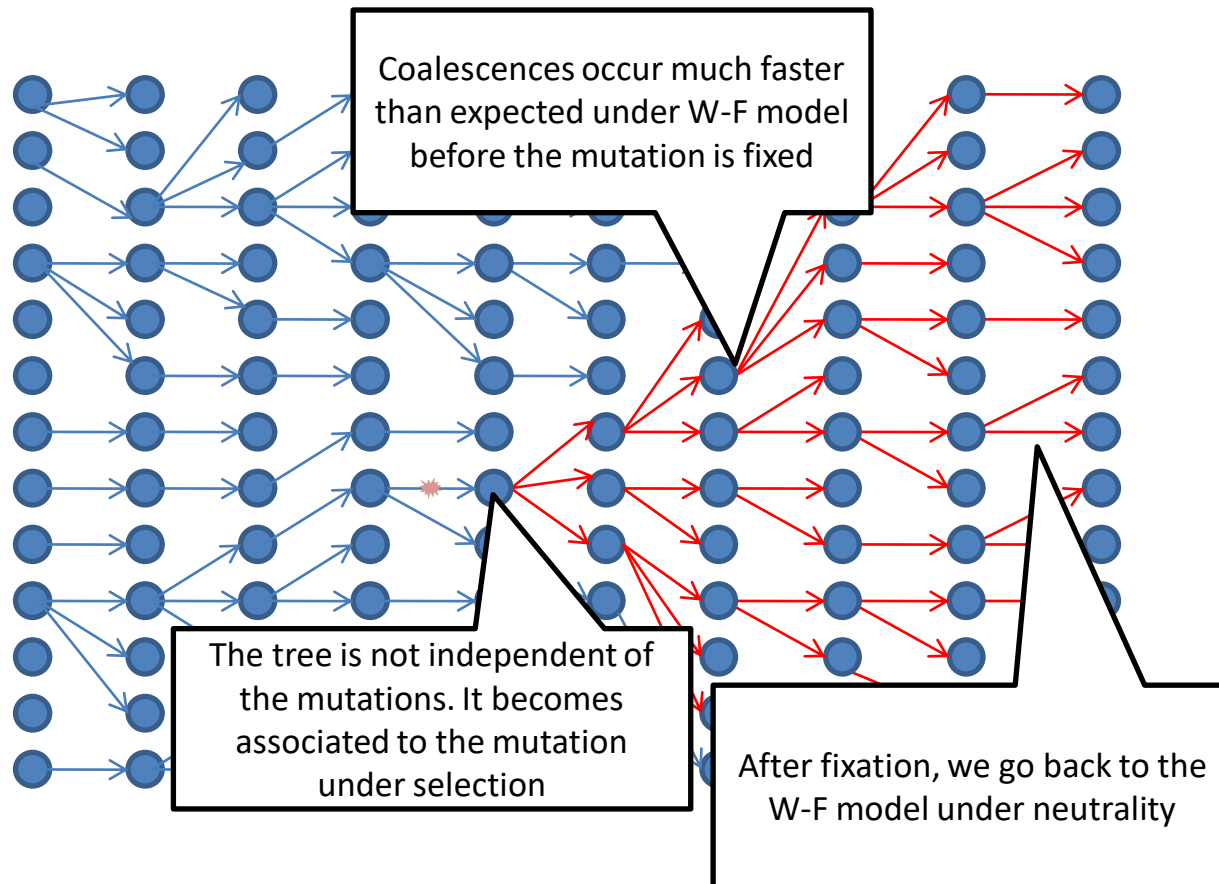


GOING BEYOND WRIGHT-FISHER POPULATION SUBSTRUCTURE AND COALESCENCE

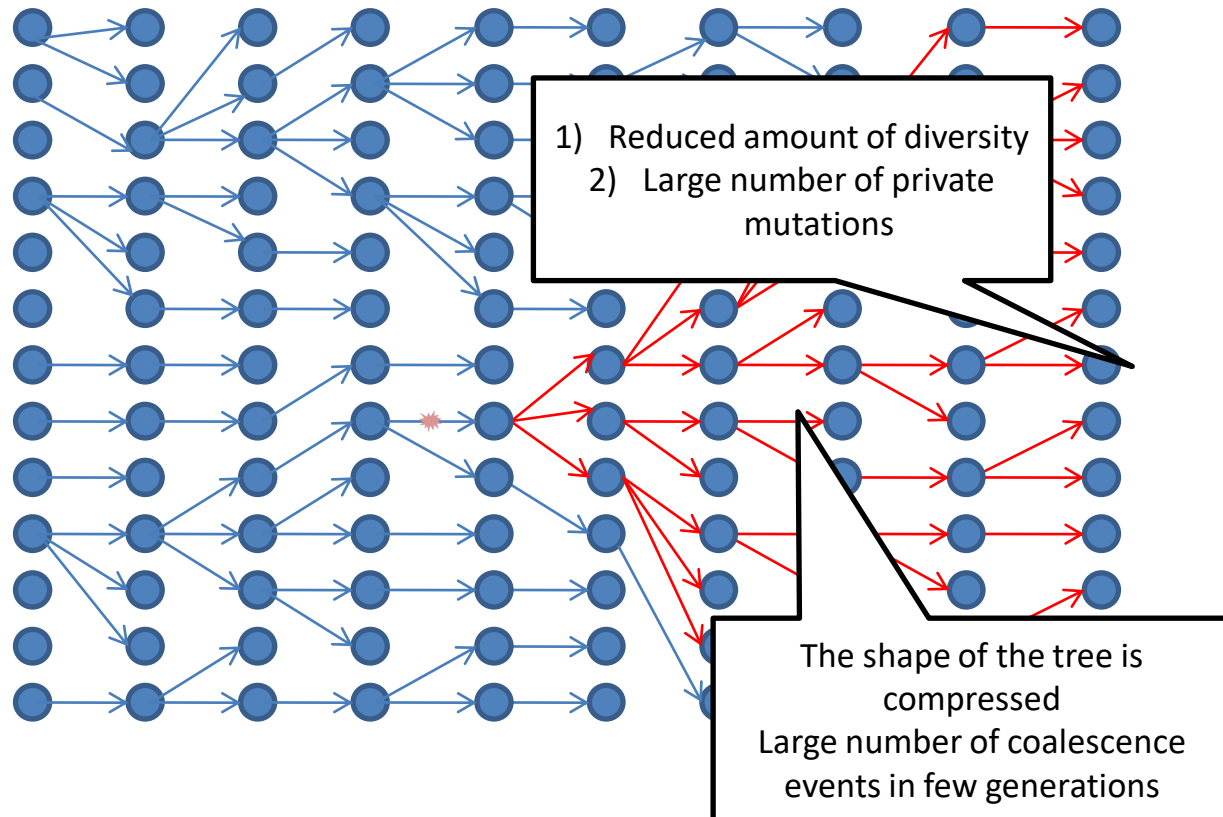
Nature Reviews Genetics volume 16, 727–740 (2015)



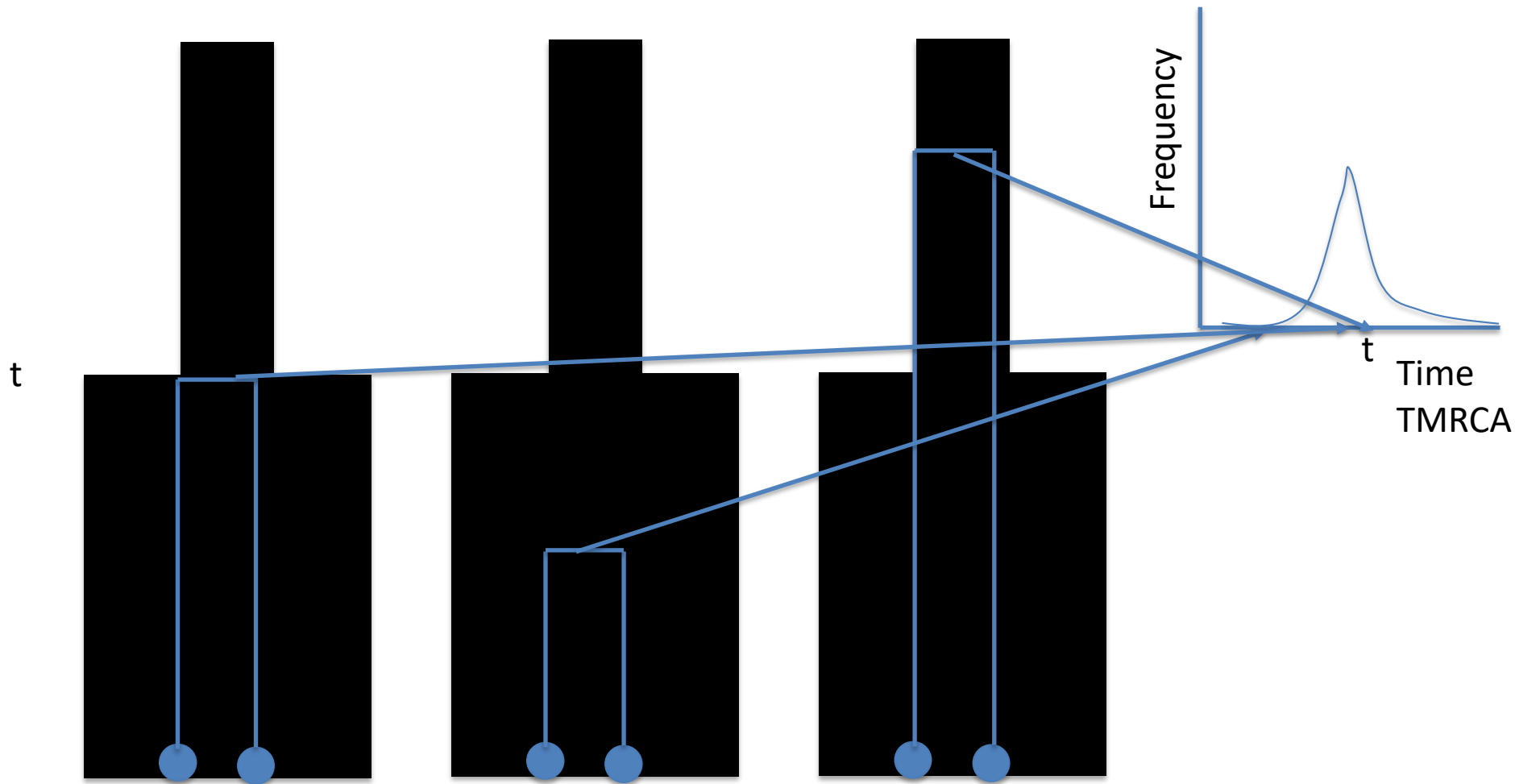
GOING BEYOND WRIGHT-FISHER SELECTION AND COALESCENCE



Coalescence under strong selection



Coalescence and inference



Examples of execution of coalescence of two chromosomes under this model

Coalescence events over time

Expected Number of Coalescent Events

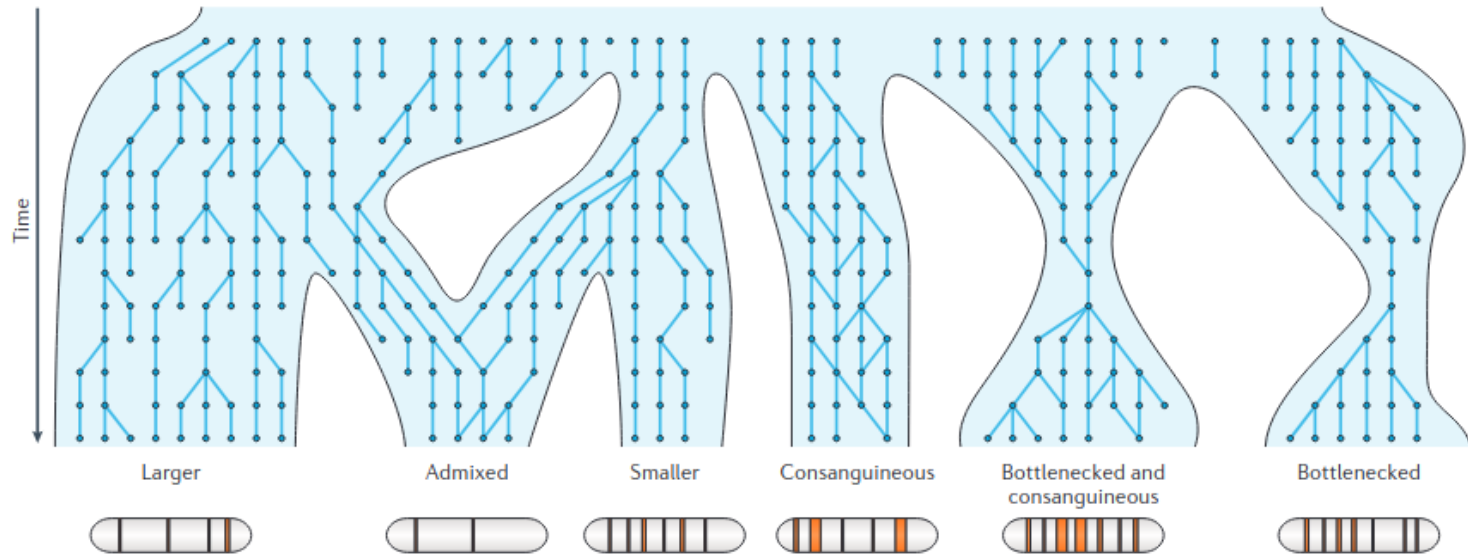
- If the effective population size $N_e(t)$ changes over a time interval $[t_1, t_2]$, the expected number of coalescent events is:

$$\mathbb{E}[\text{Coalescent Events}] = \binom{k}{2} \cdot \int_{t_1}^{t_2} \frac{1}{2N_e(t)} dt$$

- Interpretation: Sum of coalescence hazards over time interval
- Equals cumulative coalescent opportunity across all pairs of lineages
- Applies to coalescent simulators and likelihood inference of demography

Coalescence events over time

Coalescent rate of a sample depends on all demographic and evolutionary process of its genealogy



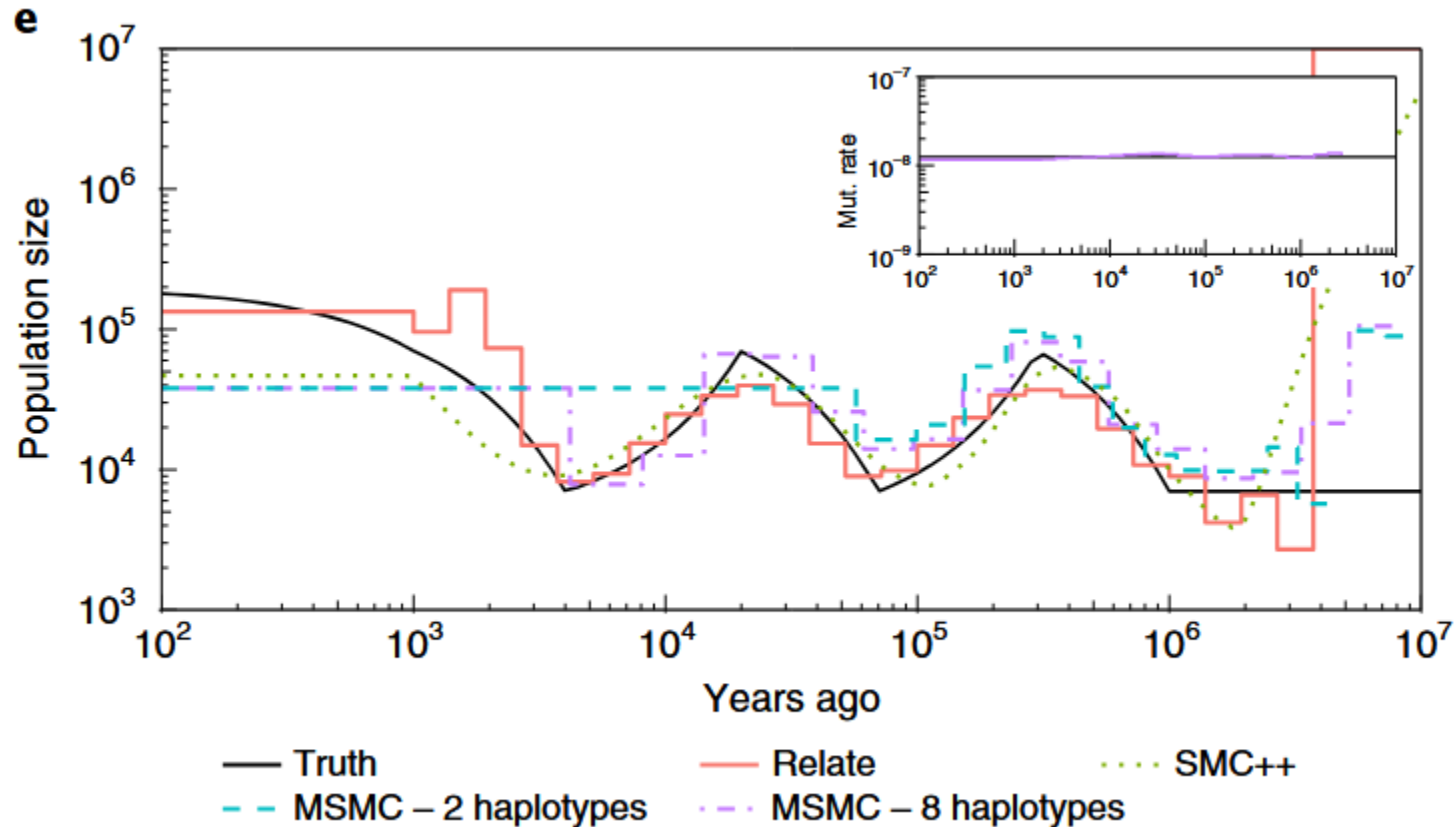
$$P(T_2 \leq t) = \frac{P(T_2 = t)}{1 - P(T_2 < t)} = \frac{pdf}{1 - cdf} = \frac{1}{IICR(t)}$$

$$IICR(t) = \frac{1}{\text{coalescence rate}(t)} = 2N_e(t) \Rightarrow N_e(t) = \frac{IICR(t)}{2}$$

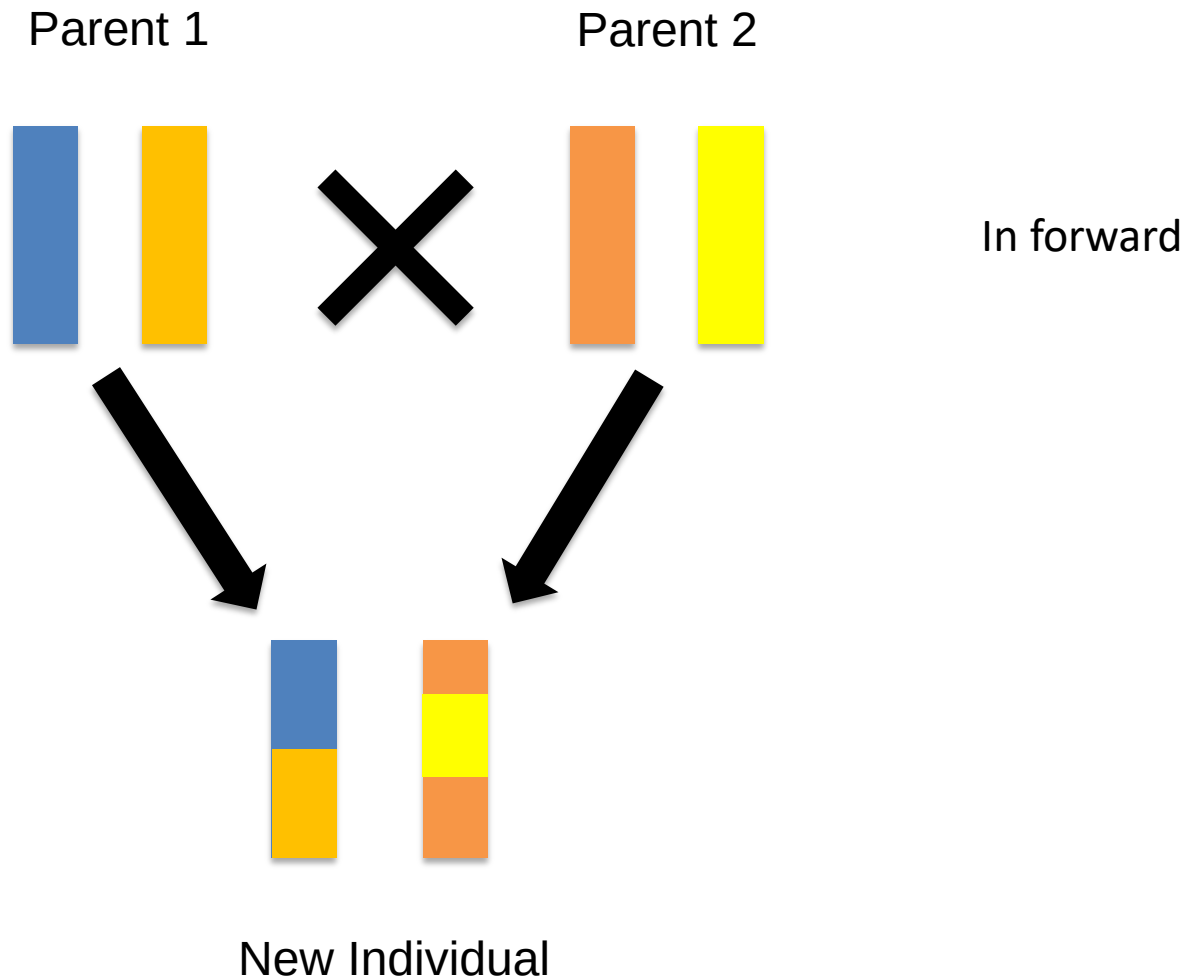
Adapted from Alba Nieto

Estimating effective population size thanks to TMCRA

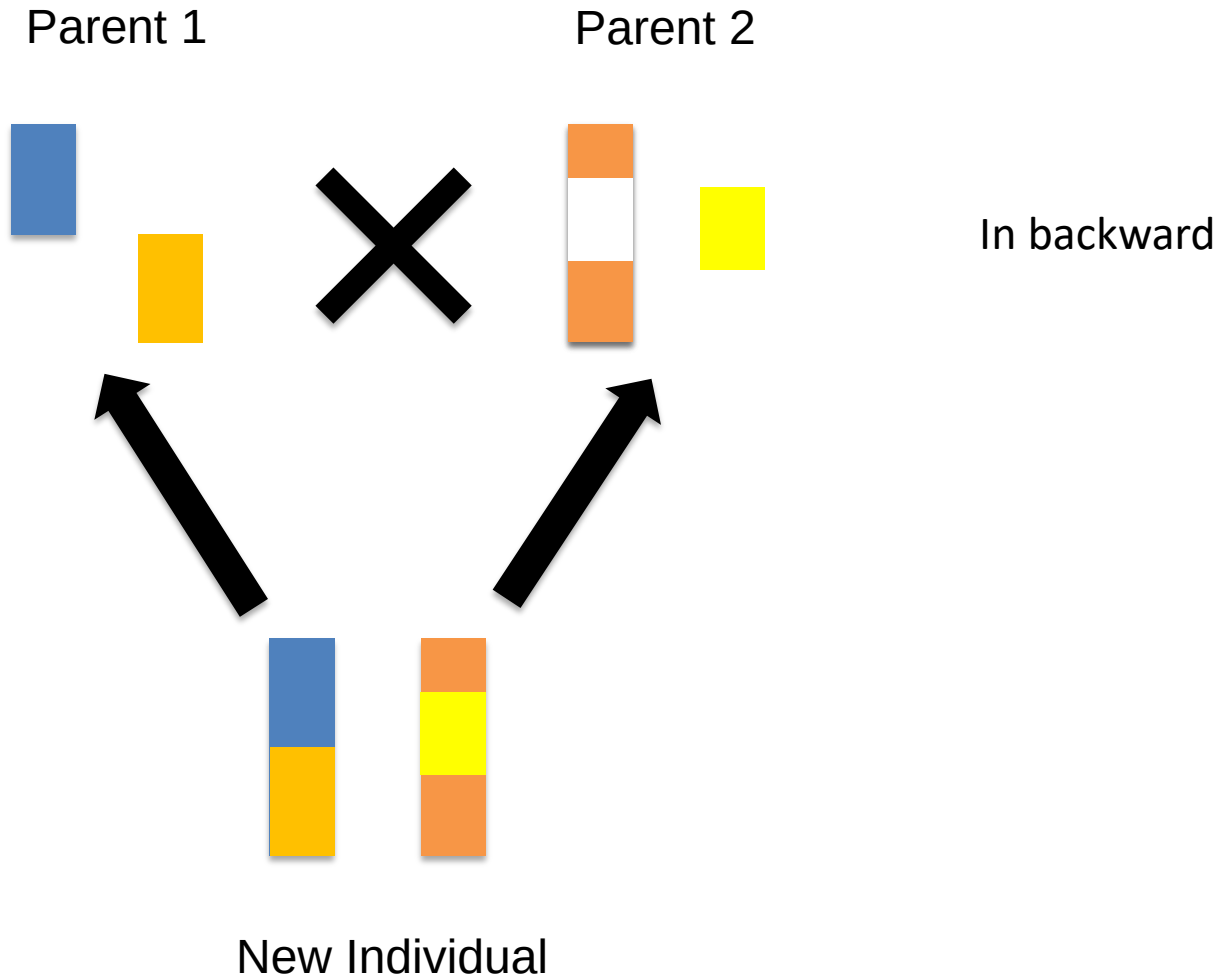
If we assume we have a single evolving population, then the coalescence rates are proxies for N_e



Diploid organisms also have RECOMBINATION

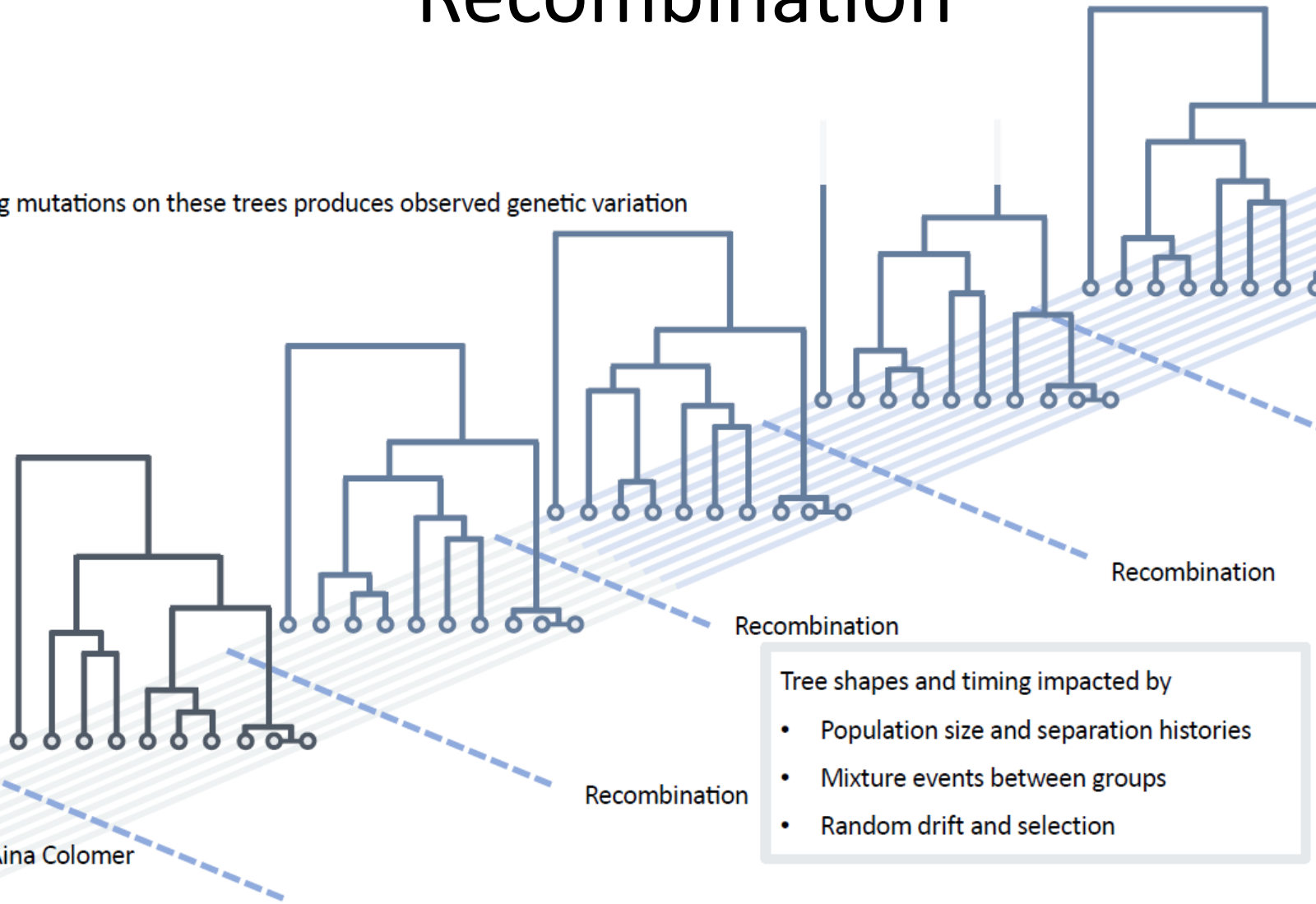


Diploid organisms also have RECOMBINATION



Modelling Coalescence with Recombination

Placing mutations on these trees produces observed genetic variation



Credit: Aina Colomer

Diploid organisms also have RECOMBINATION

The era of the ARG: an empiricist's guide to ancestral recombination graphs

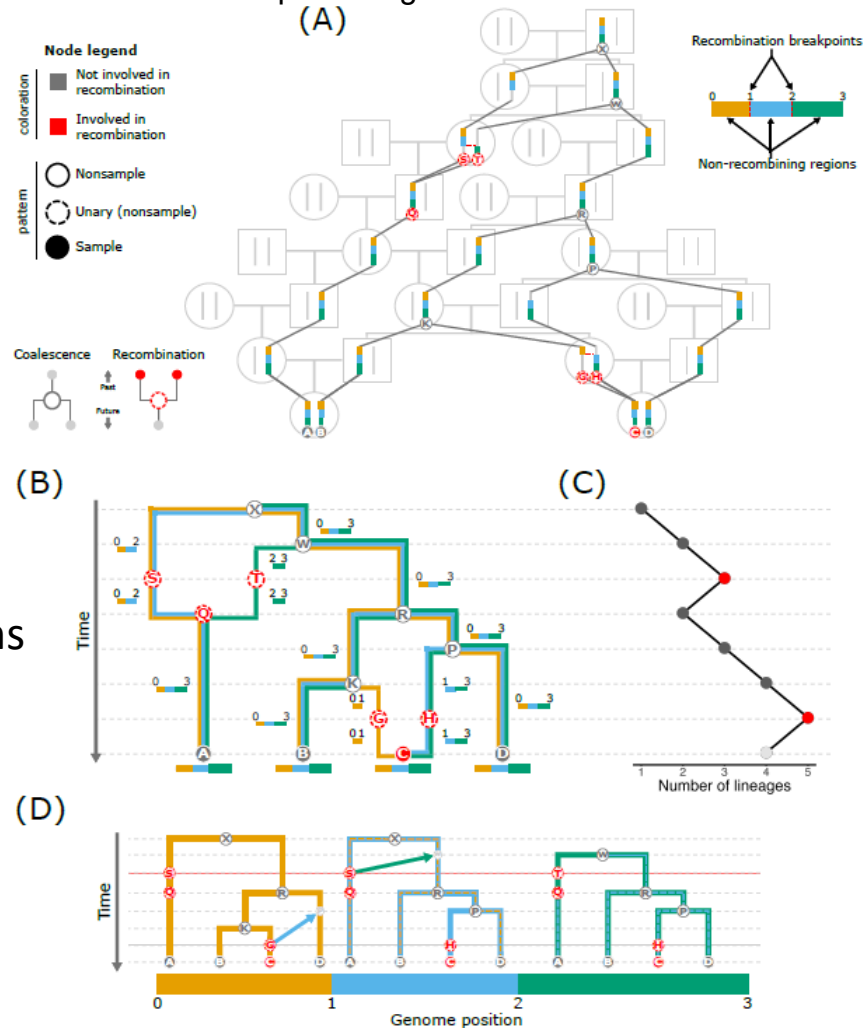
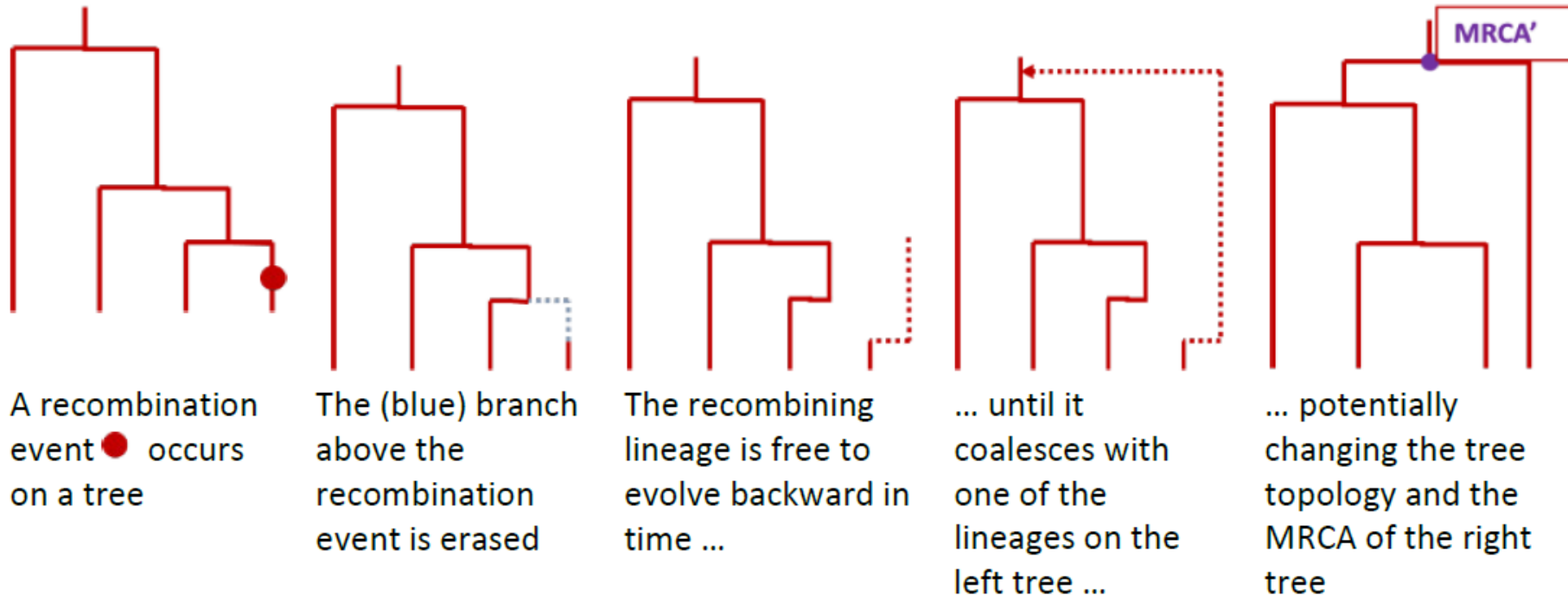


Figure 1

Diploid organisms also have RECOMBINATION

Secuential Markov Chain algorithm for simulating genomic regions with recombination

Illustration of the SMC algorithm



Diploid organisms also have RECOMBINATION

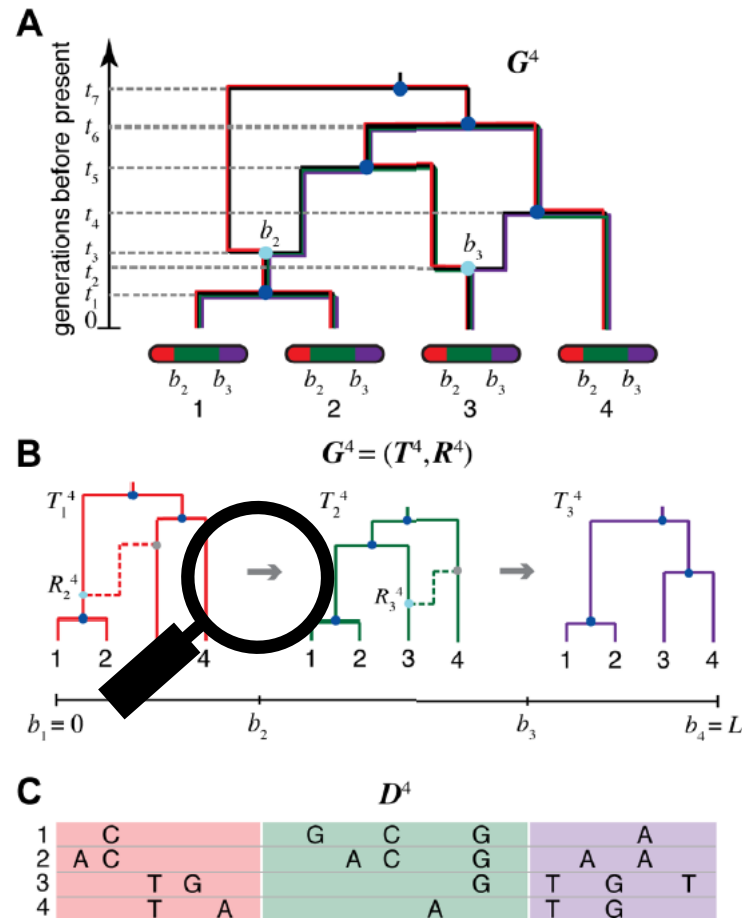
Algorithm 1: Generating genealogies under the SMC' (Marjoram and Wall 2006).

Input: Effective population size N_e , sequence length L , and per-site per-generation recombination rate r . Compute the population-scaled recombination rate $\rho = 4N_e r$.

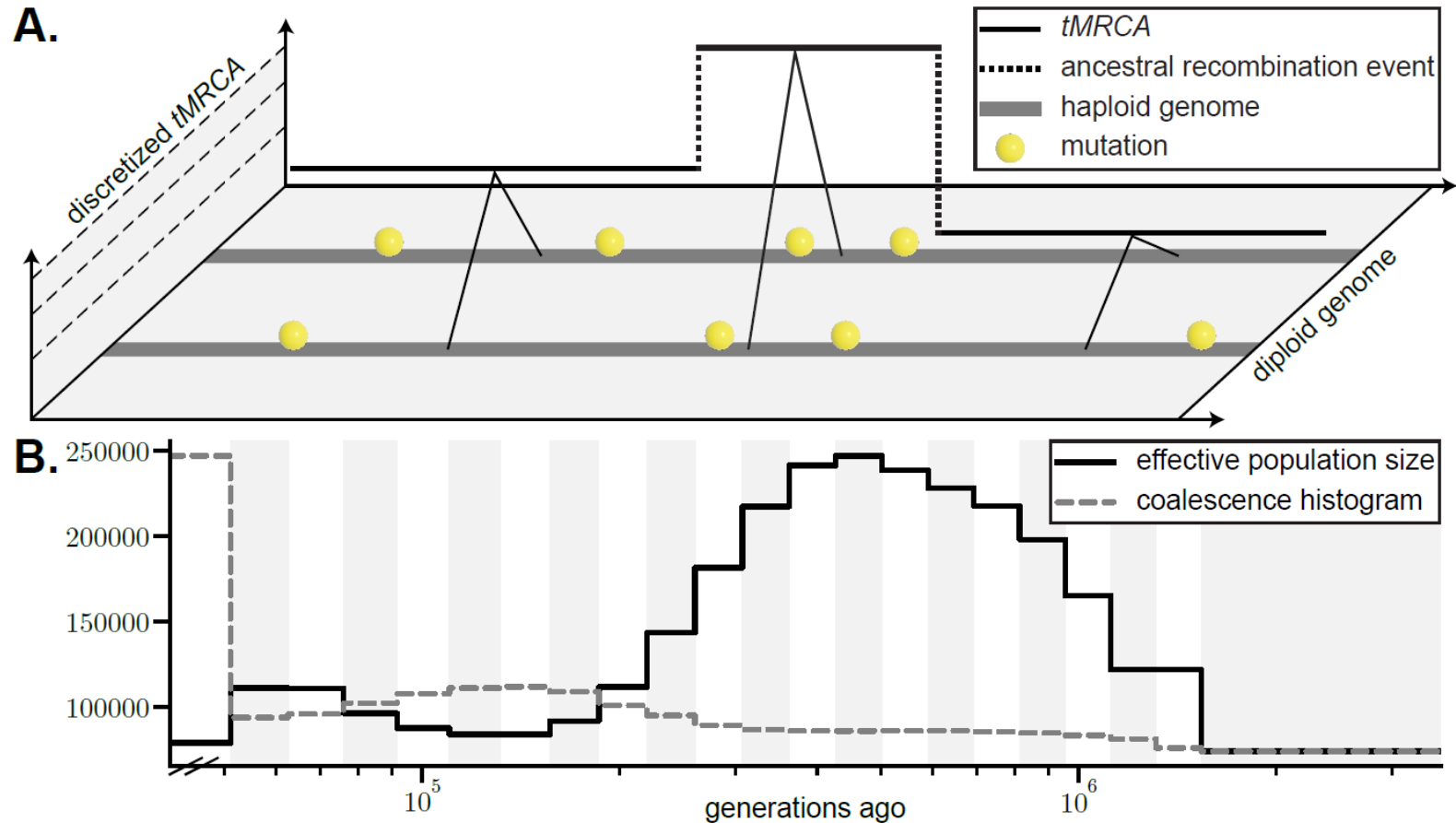
1. Set the first left interval position as $x = 0$ and generate a coalescent tree for x under Kingman's coalescent, denoted as $\mathcal{T}(x)$. Denote the length of the tree at x as $\mathcal{L}(x)$, which sums all of the branch lengths in the tree.
2. Generate the right interval position as $y \sim \text{Exp}(\frac{\rho}{2} \mathcal{L}(x))$, the distance along the chromosome until the next recombination event—i.e., tree $\mathcal{T}(x)$ spans the genomic interval $[x, x + y]$.
3. Pick a point g on the tree $\mathcal{T}(x)$ uniformly.
4. Add a recombination event to the tree $\mathcal{T}(x)$ at point g , resulting in a graph, which occurs at chromosomal location $x + y$. The left emerging branch follows the original path in $\mathcal{T}(x)$, while the right emerging branch re-coalesces with a lineage in $\mathcal{T}(x)$ at some point c higher up on the graph (after a waiting time which is exponentially distributed, with rate proportional to the number of ancestral lineages at each epoch).
5. Delete the part of the left emerging branch that lies between the newly added recombination event at point g and the next coalescence event along the branch, thus reverting the graph back to a tree.
6. Set the new left interval position as $\hat{x} = x + y$, where $\mathcal{T}(\hat{x})$ and $\mathcal{L}(\hat{x})$ denote the new tree constructed starting from position $\hat{x}$ and its associated tree length.
7. if $\hat{x} < L$ then
 Set $x = \hat{x}$ and return to step 2.

Not Just Ne Ne-more: New Applications for SMC from Ecology to Phylogenies

Remember: we have to infer it from the genomic data

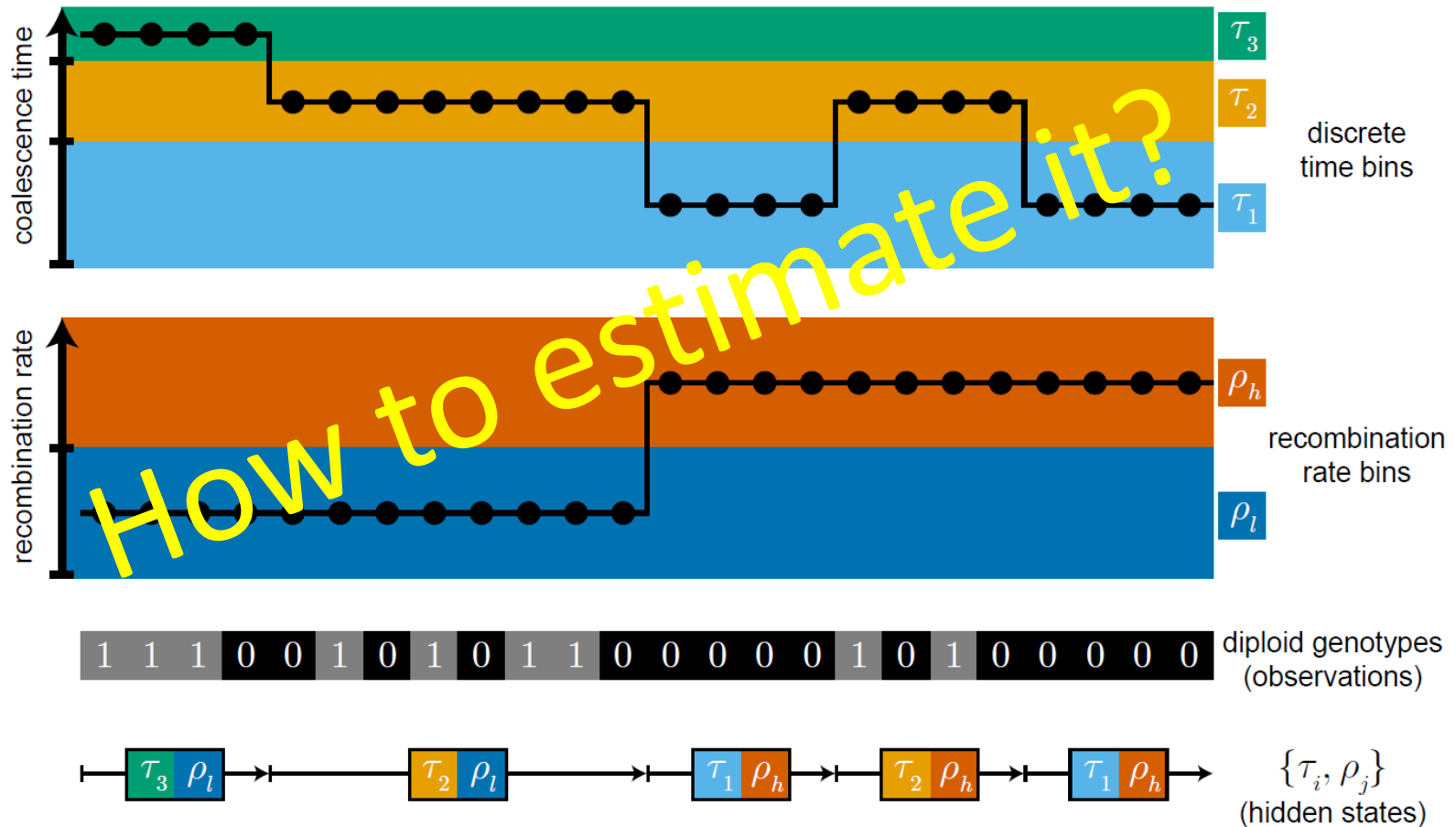


Identifying chunks with the same tMRCA



Not Just Ne Ne-more: New Applications for SMC from Ecology to Phylogenies

Remember: we have to infer it from the genomic data



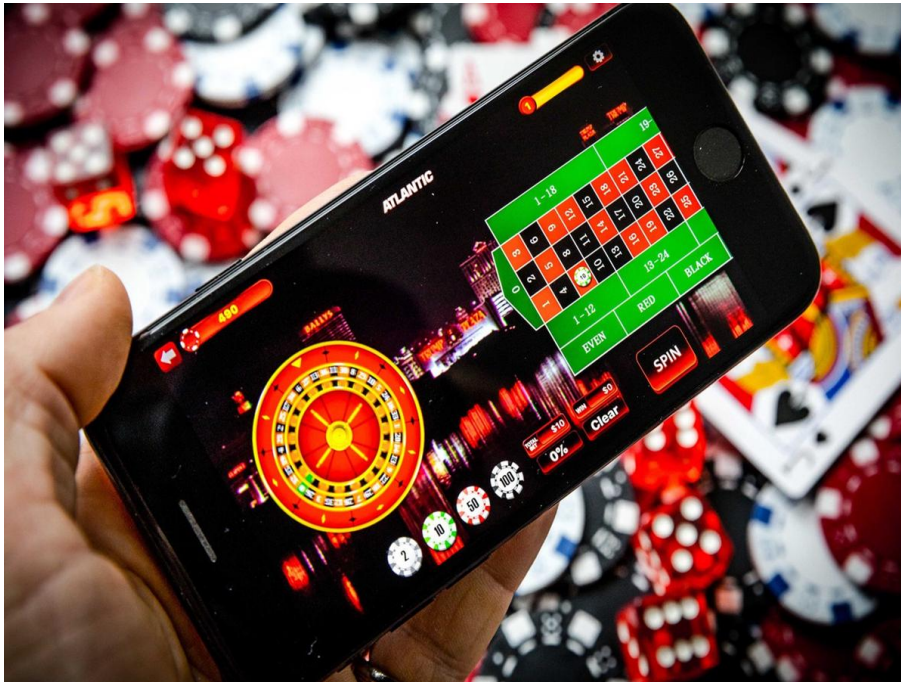
Not Just Ne Ne-more: New Applications for SMC from Ecology to Phylogenies

SMC in one slide

- hidden state = local genealogy / TMRCA interval
- transitions = recombination
- emissions = sequence variation / heterozygosity
- output = coalescence rate history → iicr

HMM: A simple working case

The occasionally dishonest casino problem

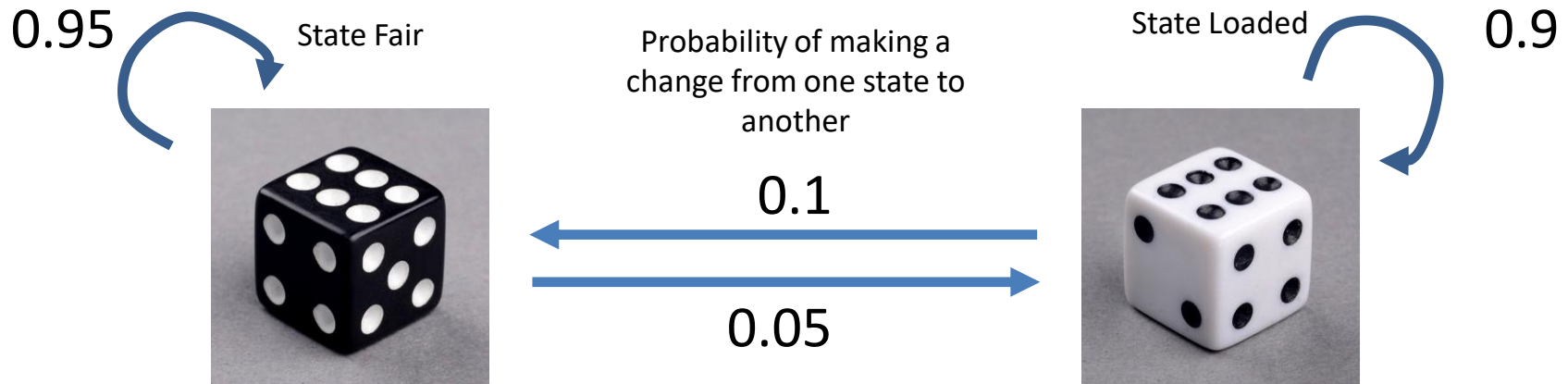


With some probability, the casino uses a dice that is **LOADED** so the number six (the bench wins) occurs more often ($P(6)=0.5$) than expected at random.

HOW CAN WE KNOW WHEN IT IS LOADED OF FAIR?

HMM: A simple working case

The occasionally dishonest casino problem



Emission probabilities of fair dice

1	2	3	4	5	6
1/6	1/6	1/6	1/6	1/6	1/6

Fair (F) dice

Emission probabilities of loaded dice

1	2	3	4	5	6
1/10	1/10	1/10	1/10	1/10	1/2

Loaded (L) dice

HMM: Probability of x given model?

$$P(1,3,4,5,1,2,6,6,3,6,6,2,1,2,4,5 \mid \text{HMM})$$

x	$P(x F)$
1	$\frac{1}{6}$
2	$\frac{1}{6}$
3	$\frac{1}{6}$
4	$\frac{1}{6}$
5	$\frac{1}{6}$
6	$\frac{1}{6}$

Emission probabilities
of Fair

Transition probabilities

$$P(C|F) = 0.01$$

$$P(F|C) = 0.1$$

x	$P(x C)$
1	$\frac{1}{10}$
2	$\frac{1}{10}$
3	$\frac{1}{10}$
4	$\frac{1}{10}$
5	$\frac{1}{10}$
6	$\frac{1}{2}$

Emission probabilities
of Charged

It refers to the values
of emission and
transition that are
used for the k
considered states

HMM: Estimating the probability of an observed sequence of events

Probability of a sequence x given that the states of the sequence are known and we have the HMM model

Imagine we observe the sequence x

$$x = \langle x_1, x_2, x_3 \rangle = \langle 6, 1, 6 \rangle$$

Assume that the prior probability of starting at one state or at the other is the same (0.5)

What would be the probability $P(x, \pi)$ if the state sequence was

$$\pi = \langle \pi_1, \pi_2, \pi_3 \rangle = \langle F, F, F \rangle$$

$$P(x, \pi) = a_{0\pi} \prod_i^N e_{\pi_i}(x_i) a_{\pi_i \pi_{i+1}}$$

$$P(x, \pi) = 0.5 * 0.95 * \frac{1}{6} * 0.95 * \frac{1}{6} * 0.95 * \frac{1}{6} = 0.001984664$$

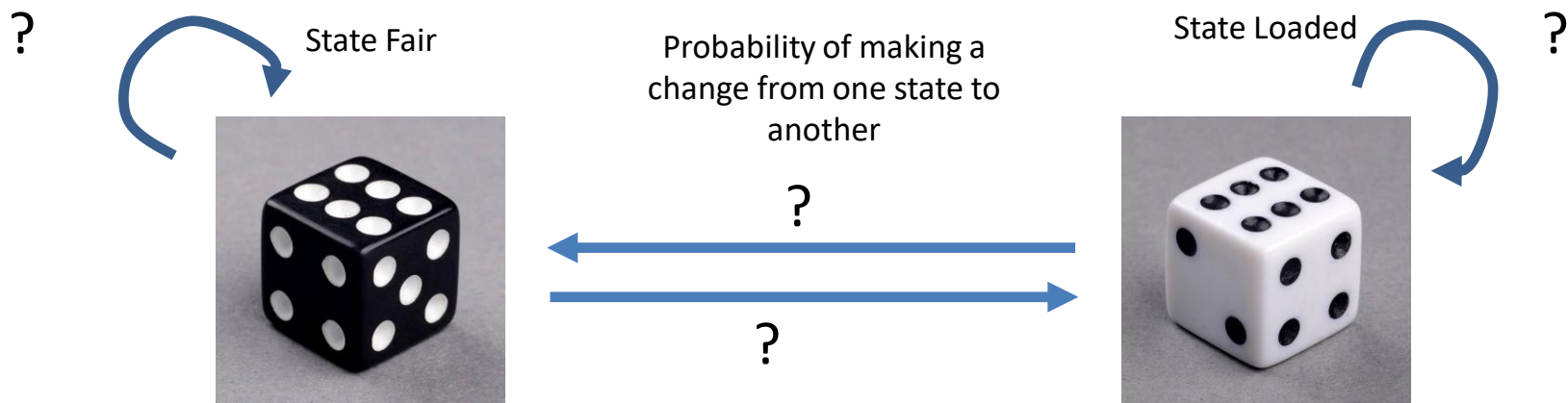
Prior of starting at F

Prob of staying at state F

Emission prob of 6 at state F

HMM

What happens when the parameters are not known?



Emission probabilities of fair dice

1	2	3	4	5	6
?	?	?	?	?	?

Fair (F) dice

Emission probabilities of loaded dice

1	2	3	4	5	6
?	?	?	?	?	?

Loaded (L) dice

$$P(1,3,4,5,1,2,6,6,3,6,6,2,1,2,4,5 | \text{HMM})$$

HMM

The occasionally dishonest casino problem

Unknown paths

x 1,4,5,1,3,2,1,6,6,4,1,6,6,6,6,6,2,3,4,5,1,4,3,6,2,1,3,5,6,6,6,6,6,6,1,2,3,1,1,1,1,2,3,4,5,5,4

$P(x|\text{HMM}=a_{rk}; e_k(b))$

Likelihood of an observed sequence given a set of transition and emission probabilities from a given HMM, computed using the forward algorithm

How to find HMM that maximizes $P(X|\text{HMM})$?

How to know at each position of the sequence which is the most likely state (i.e. tmrca class)?

HMM analogy mapped to SMC

- hidden state = TMRCA class
- observations = heterozygous / non-heterozygous sites
- transition rates depend on recombination
- emission rates depend on mutation

What the algorithm estimates

- local state posterior probabilities
- demographic parameters over time bins
- likelihood of the data under the model
- genealogies are simplified
- discretization into time intervals
- model misspecification matters

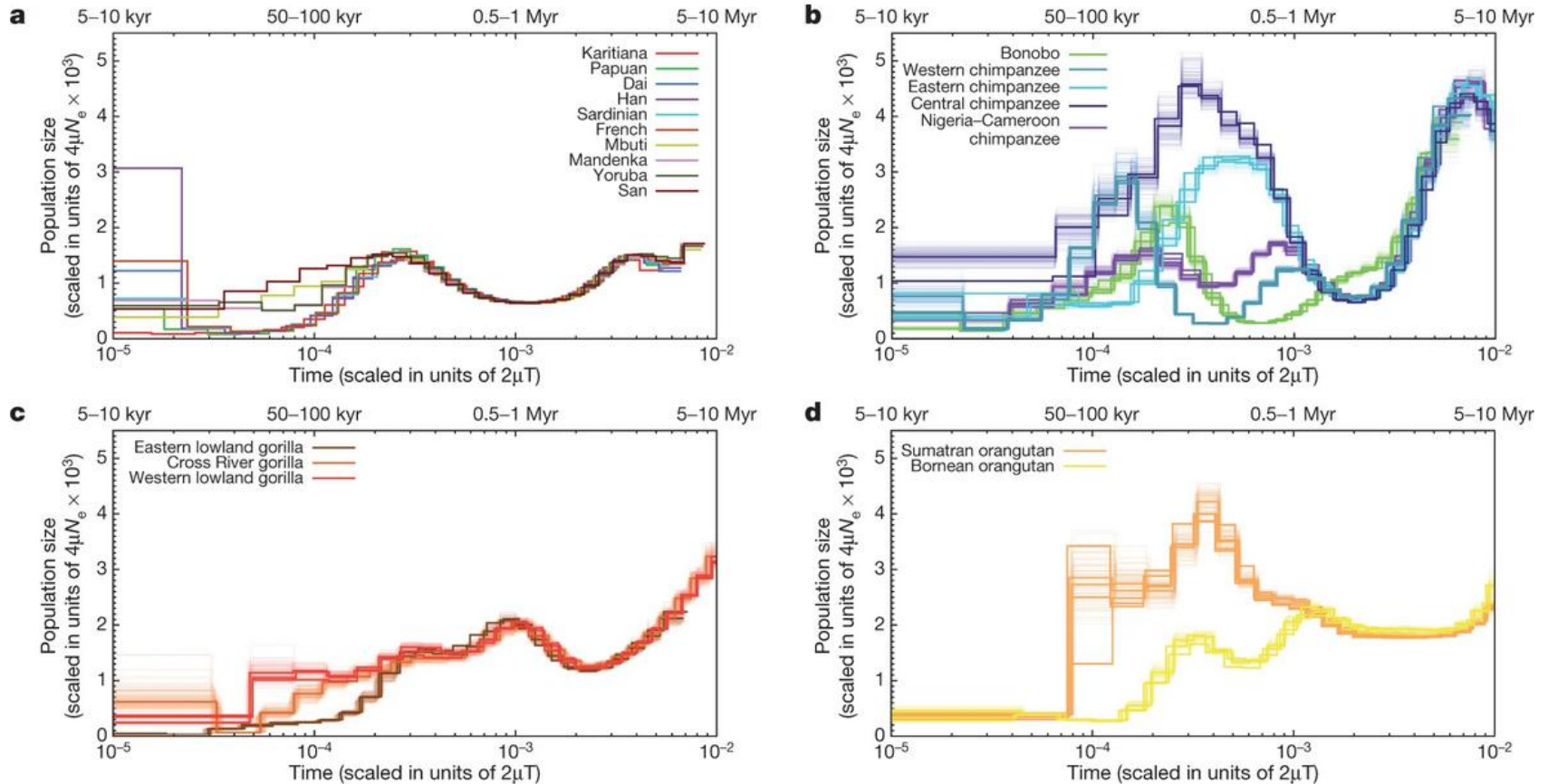
Algorithms to Infer Ne over Time

Method	Data Input	Time Scale	Assumptions	Software/Tool
PSMC	One diploid genome	Ancient (~10 kya to 1 Mya)	Constant mutation rate, recombination map	psmc
MSMC	Multiple phased haplotypes	Mid-ancient (~2 kya to 1 Mya)	Coalescent with recombination	msmc, msmc2
SMC++	Many unphased genomes + SFS	Recent to ancient	Combines SFS and HMM; scalable	smc++
Stairway Plot	Site Frequency Spectrum (SFS)	Broad (user-defined bins)	Neutral model, correct SFS	stairway_plot
Relate	Large number of phased genomes	Fine-grained, genome-wide	Ancestral recombination graph; mutation rate known	relate
GONE	Linkage disequilibrium (LD) from genotypes	Recent (0–100 generations)	Random mating, LD-based	GONE
IBDNe	Segments of identity by descent (IBD)	Recent (up to ~200 gens)	Requires accurate IBD inference	IBDNe
NeON	SFS from low coverage or pooled data	Recent (decades–centuries)	Optimized for conservation/paleogenomics	NeON R package

Why so many methods?

- one diploid genome → PSMC
- a few phased genomes → MSMC/MSMC2
- many genomes → SMC++
- very recent history → GONE / IBDNe
- explicit genealogies → Relate

Example of PSMC output

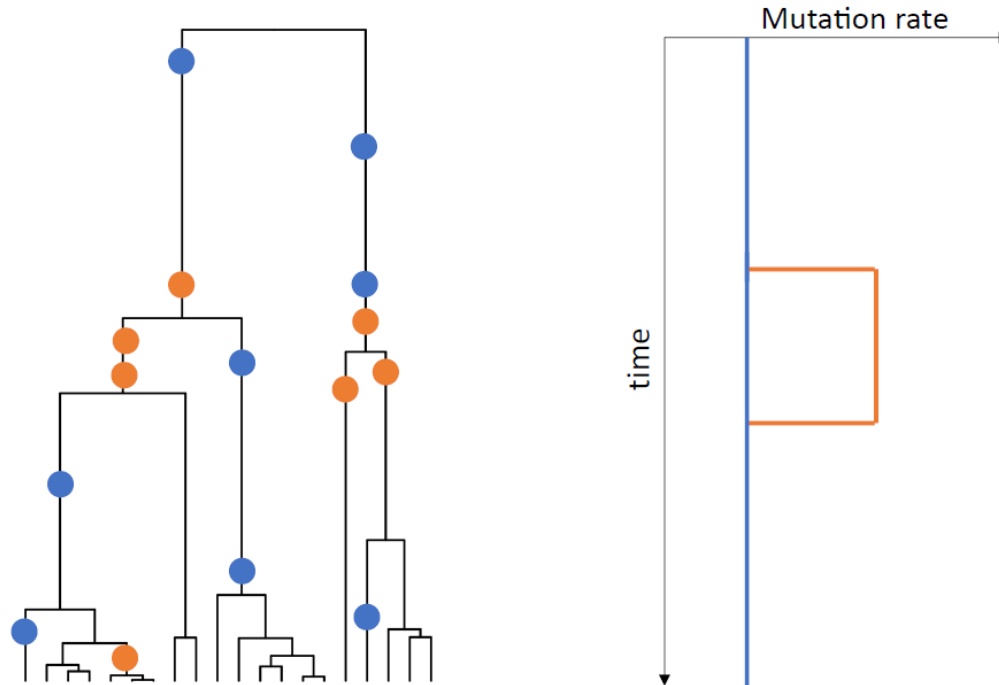


Using Relate for Genome-Wide Coalescence

- Relate infers genome-wide genealogies
- Uses thousands of genomes
- Outputs: Coalescence rates, N_e curves, tree sequences

Relate & mutation rate

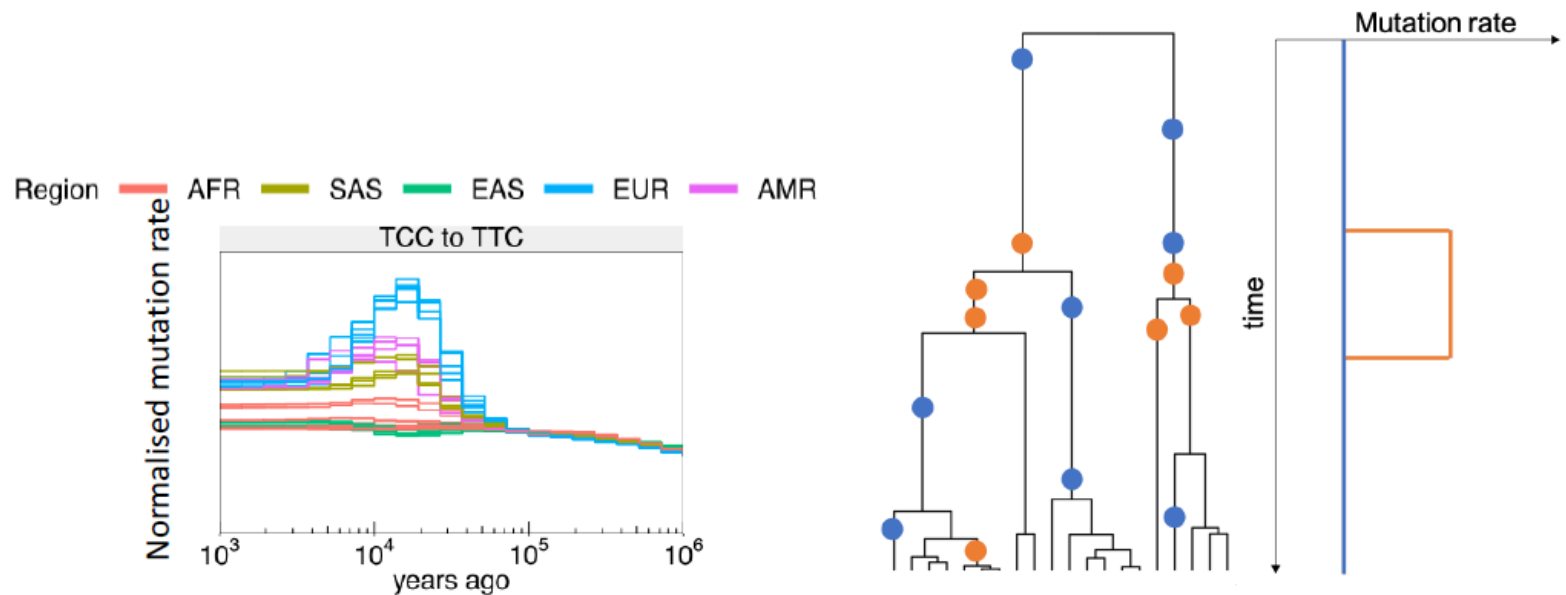
Clusters of mutations in time can capture changes in mutation rate



from Leo Speidel

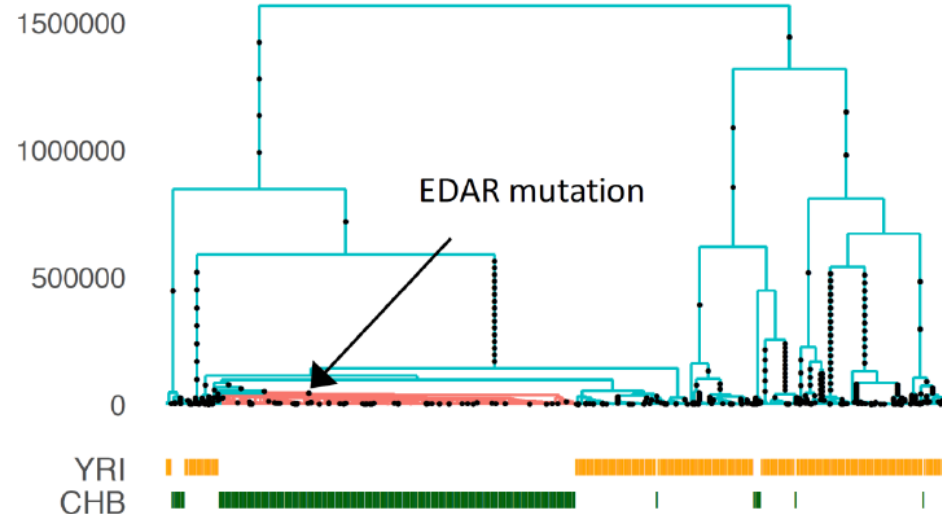
Relate & mutation rate

TCC/TTC mutation rates have experienced a strong pulse in the Upper Paleolithic



How did this spread to all West Eurasians today?

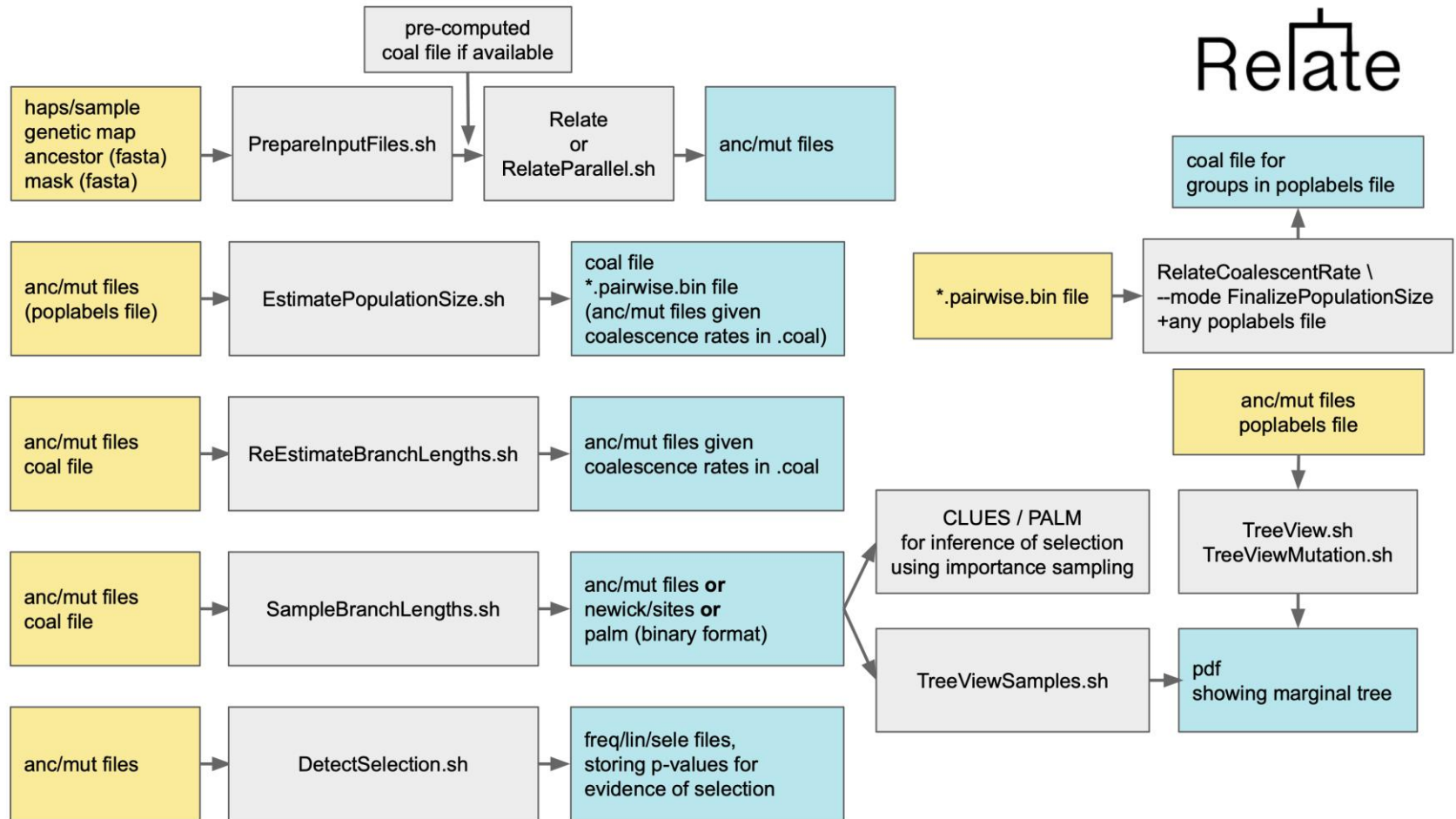
Relate & positive selection



Quantifying positive natural selection on a single mutation

- Genetic adaptations to changing environment, diet, lifestyles,...
- Use trees incorporating demographic history

RELATE



Limits of SMC-based models

Limits and convergence properties of the sequentially Markovian coalescent

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München

Abstract

Several methods based on the sequentially Markovian coalescent (SMC) make use of full genome sequence data from samples to infer population demographic history including past changes in population size, admixture, migration events and population structure. More recently, the original theoretical framework has been extended to allow the simultaneous estimation of population size changes along with other life history traits such as selfing or seed banking. The latter developments enhance the applicability of SMC methods to nonmodel species. Although convergence proofs have been given using simulated data in a few specific cases, an in-depth investigation of the limitations of SMC methods is lacking. In order to explore such limits, we first develop a tool inferring the best case convergence of SMC methods assuming the true underlying coalescent genealogies are known. This tool can be used to quantify the amount and type of information that can be confidently retrieved from given data sets

- Presence of transposons or repetitive elements
- Extreme demographies
- High recombination and mutation rates

Limits of SMC-based models

Accurate inference of population history in the presence of background selection

 Trevor Cousins, Daniel Tabin, Nick Patterson, David Reich, Arun Durvasula

doi: <https://doi.org/10.1101/2024.01.18.576291>

This article is a preprint and has not been certified by peer review [what does this mean?].

Abstract

Full Text

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Metrics

 Preview PDF

1 Abstract

All published methods for learning about demographic history make the simplifying assumption that the genome evolves neutrally, and do not seek to account for the effects of natural selection on patterns of variation. This is a major concern, as ample work has demonstrated the pervasive effects of natural selection and in particular background selection (BGS) on patterns of genetic variation in diverse species. Simulations and theoretical work have shown that methods to infer changes in effective population size over time ($N_e(t)$) become increasingly inaccurate as the strength of linked selection increases. Here, we introduce an extension to the Pairwise Sequentially Markovian Coalescent (PSMC) algorithm, PSMC+, which explicitly co-models demographic history and natural selection. We benchmark our method using forward-in-time simulations with BGS and find that our approach improves the accuracy of effective population size inference. Leveraging a high resolution map of BGS in humans, we infer

- Background selection

Limits of SMC-based models

PSMC analysis of effective population sizes in molecular ecology and its application to black-and-white *Ficedula* flycatchers

[Krystyna Nadachowska-Brzyska](#)^{1,✉}, [Reto Burri](#)¹, [Linnéa Smeds](#)¹, [Hans Ellegren](#)^{1,✉}

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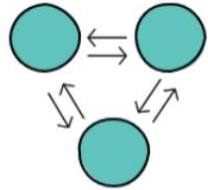
PMCID: PMC4793928 PMID: [26797914](#)

Abstract

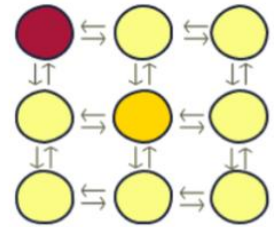
Climatic fluctuations during the Quaternary period governed the demography of species and contributed to population differentiation and ultimately speciation. Studies of these past processes have previously been hindered by a lack of means and genetic data to model changes in effective population size (N_e) through time. However, based on diploid genome sequences of high quality, the recently developed pairwise sequentially Markovian

- Variant calling restrictions (quality)

WARNING: ASSUMPTIONS!!!!

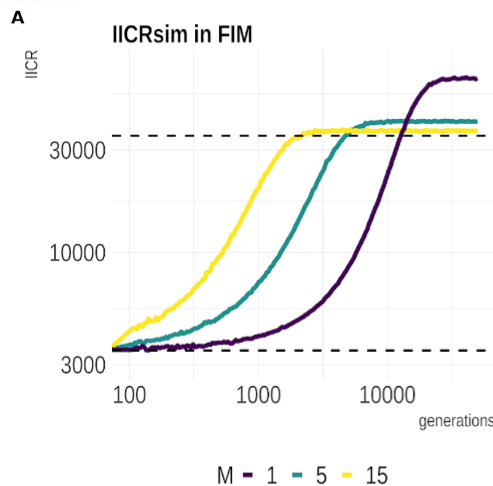


Expected IICR under population structure

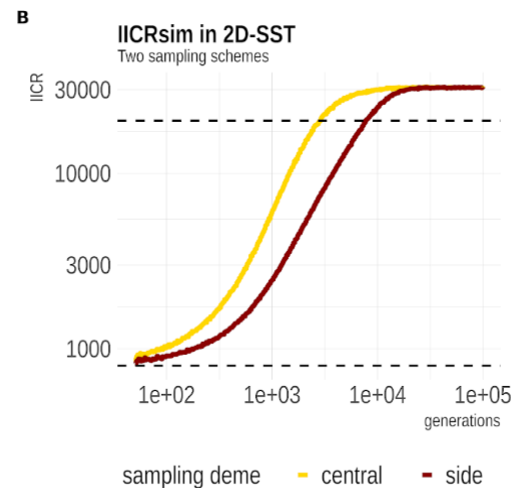


2-dimensional stepping stone (2D-SST)

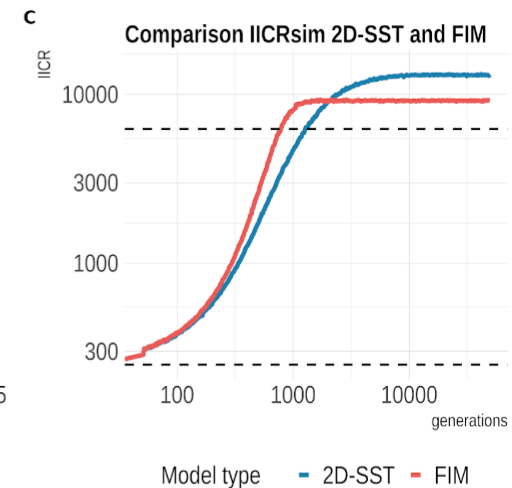
Finite Islands Model (FIM)



Demographic parameters

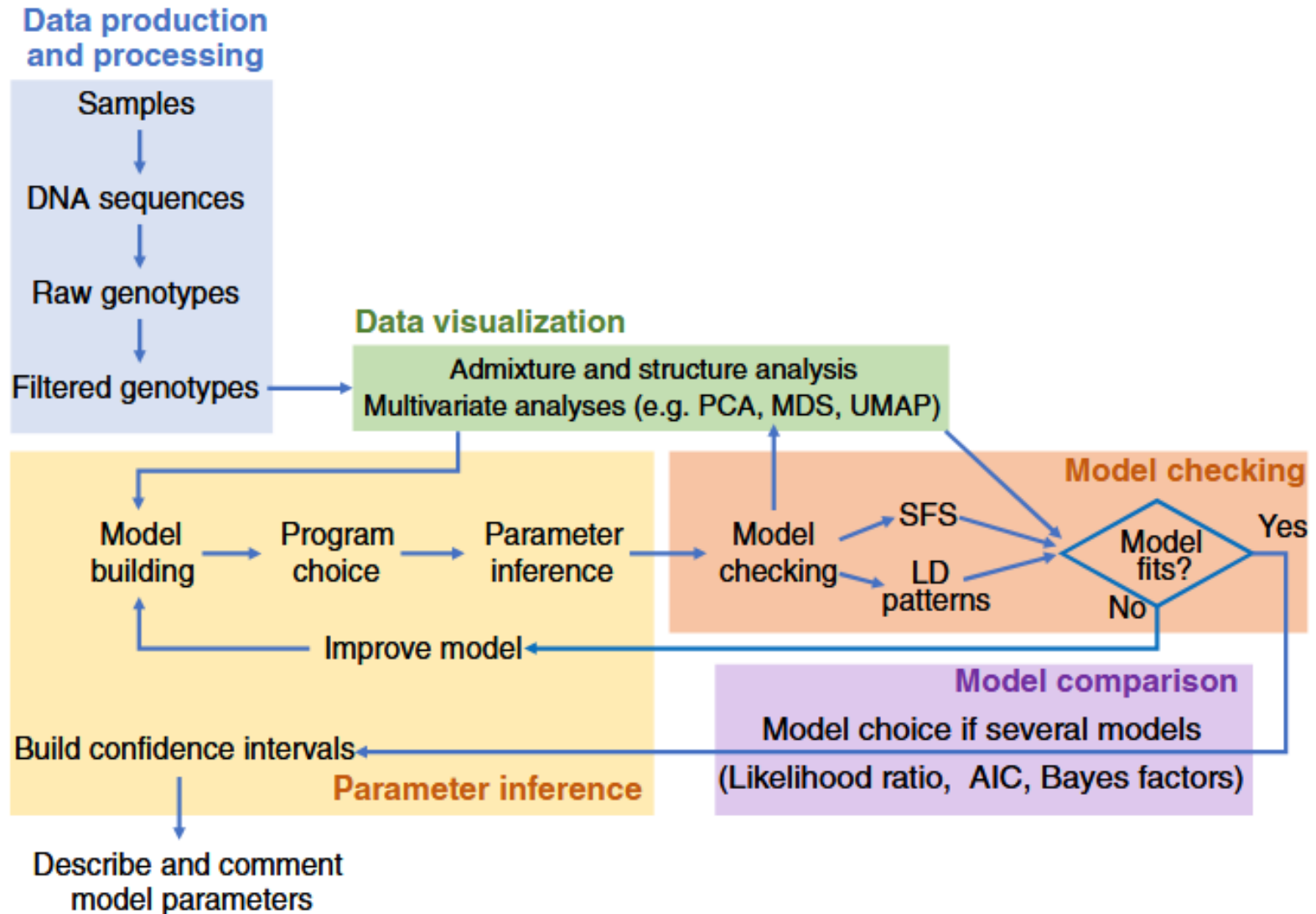


Sampling sites



Type of underlying model

Demographic modelling



Best-practice workflow

- define the biological question
- choose the right method for the time scale and data type
- apply strict masking and QC
- calibrate with mutation rate and generation time
- quantify uncertainty
- validate with simulations or alternative summaries
- interpret conservatively

More on SMC-based models

 > q-bio > arXiv:2506.00692

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Quantitative Biology > Populations and Evolution

[Submitted on 31 May 2025]

Not Just N_e N_e -more: New Applications for SMC from Ecology to Phylogenies

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Genomes contain the mutational footprint of an organism's evolutionary history, shaped by diverse forces including ecological factors, selective pressures, and life history traits. The sequentially Markovian coalescent (SMC) is a versatile and tractable model for the genetic genealogy of a sample of genomes, which captures this shared history. Methods that utilize the SMC, such as PSMC and MSMC, have been widely used in evolution and ecology to infer demographic histories. However, these methods ignore common biological features, such as gene flow events and structural variation. Recently, there have been several advancements that widen the applicability of SMC-based methods: inclusion of an isolation with migration model, integration with the multi-species coalescent, incorporation of ecological variables (such as selfing and dormancy), inference of dispersal rates, and many computational advances in applying these models to data. We give an overview of the SMC model and its various recent extensions, discuss examples of biological discoveries through SMC-based inference, and comment on the assumptions, benefits and drawbacks of various methods.

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SMC

- all SMC methods infer history through local genealogical changes
- recombination makes the genome informative about many coalescent events
- hidden-state models make this computationally feasible
- method choice depends on data and time scale
- interpretation requires caution because structure, selection, and technical bias can mimic demography

Thank you very much!

